

THE GIT OF $4 \times 3 \times 3$ TENSORS (DETERMINANTAL CUBIC SURFACES)

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ABSTRACT. A general $4 \times 3 \times 3$ tensor defines three determinantal varieties in $\mathbb{P}^3$ and $\mathbb{P}^2$ and $\mathbb{P}^2$ where it drops rank: a cubic surface and two length 6 subschemes. We study the Geometric Invariant Theory (GIT) quotient of $\mathbb{P}(\mathbb{C}^4 \otimes \mathbb{C}^3 \otimes \mathbb{C}^3)$ by $\mathrm{SL}(4) \times \mathrm{SL}(3) \times \mathrm{SL}(3)$, a moduli space of linear determinantal representations of cubic surfaces. We give a geometric description of the (semi)stable locus and some of the geometry of this GIT quotient. This GIT quotient admits a surjective rational map to the GIT moduli space of cubic surfaces, undefined at only one point.

1. INTRODUCTION

The determinant of a $d \times d$ matrix of linear forms in n variables is a homogeneous degree d polynomial. Going backwards, one can ask whether a given homogeneous degree d polynomial in n variables can be written as the determinant of a $d \times d$ matrix of linear forms (we call this matrix a *determinantal representation*). A quick dimension count shows that for most values of (n, d) a general polynomial has either zero or infinitely many determinantal representations (apart from the trivial case of $d \neq 1$). This is except for when $(n, d) = (3, 2), (4, 2), (4, 3)$, in which case a general polynomial has *finitely many* determinantal representations (up to $\mathrm{GL}(d) \times \mathrm{GL}(d)$ -equivalence). In the case that $(n, d) = (4, 3)$, that is of cubic surfaces in $\mathbb{P}^3$, the equation defining a sufficiently general (smooth) cubic surface can be written as the determinant of a 3×3 matrix of linear forms *in exactly 72 ways* (up to $\mathrm{GL}(3) \times \mathrm{GL}(3)$ -equivalence). A 3×3 matrix of linear forms in 4 variables is adjoint to a $4 \times 3 \times 3$ tensor. A $4 \times 3 \times 3$ tensor corresponds to a cubic surface and two length 6 subschemes of $\mathbb{P}^2$. We briefly describe this correspondence.

Let U and V and W be complex vector spaces with $\dim U = 4$ and $\dim V = \dim W = 3$. We let $\varphi \in U^\vee \otimes V^\vee \otimes W^\vee$ be a sufficiently general tensor. The tensor φ has two adjoint matrices $\varphi_U \in U^\vee \otimes \mathrm{Hom}(W, V^\vee)$ and $\varphi_V \in V^\vee \otimes \mathrm{Hom}(W, U^\vee)$. The determinant of the 3×3 matrix φ_U is a cubic polynomial $\det(\varphi_U)$, which cuts out a smooth cubic surface $S_\varphi \subset \mathbb{P}U$. The 4×3 matrix φ_V presents a free resolution

$$(1) \quad 0 \longrightarrow W \otimes \mathcal{O}_{\mathbb{P}V}(-4) \xrightarrow{\varphi_V} U^\vee \otimes \mathcal{O}_{\mathbb{P}V}(-3) \longrightarrow \mathcal{I}_{X_\varphi} \longrightarrow 0$$

of an ideal sheaf $\mathcal{I}_{X_\varphi} \subset \mathcal{O}_{\mathbb{P}V}$, which cuts out a length 6 subscheme $X_\varphi \subset \mathbb{P}V$. Twisting and taking cohomology of (1) gives an identification of $H^0(\mathbb{P}V, \mathcal{I}_{X_\varphi}(3)) \cong U^\vee$; this linear system induces a rational map $f : \mathbb{P}V \dashrightarrow \mathbb{P}U = |\mathcal{I}_{X_\varphi}(3)|$. The image of f is contained in S_φ and the map $f : \mathbb{P}V \dashrightarrow S_\varphi$ is birational. Its inverse extends to a morphism $f^{-1} : S_\varphi \rightarrow \mathbb{P}V$ which identifies the surface S_φ as the blowup of $\mathbb{P}V$ at 6 points $X_\varphi \subset \mathbb{P}V$. The inverse map f^{-1} is also induced from the tensor φ : we have an exact sequence

$$0 \longrightarrow W \otimes \mathcal{O}_{S_\varphi}(-1) \xrightarrow{\varphi_U} V^\vee \otimes \mathcal{O}_{S_\varphi} \longrightarrow \mathrm{coker}(\varphi_U) \longrightarrow 0.$$

Taking cohomology identifies $H^0(S_\varphi, \mathrm{coker}(\varphi_U)) \cong V^\vee$ and $\mathrm{coker}(\varphi_U) = (f^{-1})^* \mathcal{O}_{\mathbb{P}V}(1)$.

Conversely, let $S \subset \mathbb{P}U$ be a smooth cubic surface and let $g : S \rightarrow \mathbb{P}V$ a birational morphism. The map g is the blowup of $\mathbb{P}V$ at 6 points $X \subset \mathbb{P}V$. By the Hilbert-Burch theorem, the ideal sheaf $\mathcal{I}_X \subset \mathcal{O}_{\mathbb{P}V}$ cutting out $X \subset \mathbb{P}V$ has a free resolution of the form of (1) that is presented by a 4×3 matrix $\varphi_V \in V^\vee \otimes \text{Hom}(W, U^\vee)$ of linear forms in $V^\vee$. The determinant of its adjoint 3×3 matrix $\varphi_U \in U^\vee \otimes \text{Hom}(W, V^\vee)$ cuts out the surface $S \subset \mathbb{P}U$ (up to a $\text{GL}(U)$ -action). The 72 different birational morphisms $g : S \rightarrow \mathbb{P}V$ are in bijection with the 72 different ways to write the equation cutting out $S \subset \mathbb{P}U$ as the determinant of a 3×3 matrix valued in $U^\vee$.

The other adjoint matrix $\varphi_W \in W^\vee \otimes \text{Hom}(V, U^\vee)$ presents a free resolution

$$(2) \quad 0 \longrightarrow V \otimes \mathcal{O}_{\mathbb{P}W}(-4) \xrightarrow{\varphi_W} U^\vee \otimes \mathcal{O}_{\mathbb{P}W}(-3) \longrightarrow \mathcal{I}_{Y_\varphi} \longrightarrow 0$$

of an ideal sheaf $\mathcal{I}_{Y_\varphi} \subset \mathcal{O}_{\mathbb{P}W}$ in the same way as (2), which cuts out 6 points $Y_\varphi \subset \mathbb{P}W$. The surface S_φ is also isomorphic to the blowup $\text{Bl}_{Y_\varphi} \mathbb{P}W$. The pair of point configurations $X_\varphi \subset \mathbb{P}V$ and $Y_\varphi \subset \mathbb{P}W$ are Gale-dual, in the sense of [EP00]. This correspondence induces an isomorphism between the moduli spaces of:

- (a) triples $([S], [X], [Y]) \in \text{Hilb}_{\text{cubic surfaces}}(\mathbb{P}U) \times \text{Hilb}_6(\mathbb{P}V) \times \text{Hilb}_6(\mathbb{P}W)$ where $X \subset \mathbb{P}V$ and $Y \subset \mathbb{P}W$ are a pair of Gale-dual point configurations and S is a smooth cubic surface isomorphic to both $\text{Bl}_X \mathbb{P}V$ and $\text{Bl}_Y \mathbb{P}W$; and
- (b) tensors $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ where $\det(\varphi_U)$ cuts out a smooth cubic surface $S_\varphi \subset \mathbb{P}U$.

This correspondence is extremely well studied. In particular, it extends to triples (S, X, Y) consisting of a cubic surface $S \subset \mathbb{P}U$ with at worst canonical singularities and an associated pair of Gale-dual length 6 curvilinear subschemes $X \subset \mathbb{P}V$ and $Y \subset \mathbb{P}W$ to the surface S , provided that the subschemes X and Y are also not contained in a conic.

Quotienting by $\text{SL}(V) \times \text{SL}(W)$ gives a compactification $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee) // (\text{SL}(V) \times \text{SL}(W))$ of the moduli space of determinantal representations of cubic surfaces. This GIT quotient is closely related to the Hilbert schemes of twisted cubics inside cubic surfaces and has been extensively studied in [LLSS17].

Quotienting by $\text{SL}(U) \times \text{SL}(W)$ gives a compactification $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee) // (\text{SL}(U) \times \text{SL}(W))$ of the moduli space of 6 unordered points in $\mathbb{P}V$. By [ABCH13], this GIT quotient is the final minimal model of the Hilbert scheme $\text{Hilb}_6(\mathbb{P}V)$ and the birational contraction

$$\text{Hilb}_6(\mathbb{P}V) \dashrightarrow \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee) // (\text{SL}(U) \times \text{SL}(W))$$

contracts the divisor parametrising length 6 subschemes $X \subset \mathbb{P}V$ contained in a conic.

This paper is concerned with the triple quotient of $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ by the group $G := \text{SL}(U) \times \text{SL}(V) \times \text{SL}(W)$; the GIT compactification of the moduli problems of (a) and (b) after forgetting the bases of all three vector spaces U and V and W . The rational map

$$\det : \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee) \dashrightarrow |\mathcal{O}_{\mathbb{P}U}(3)|$$

sending a $4 \times 3 \times 3$ tensor to its determinant cubic descends to a rational map of GIT quotients

$$\det_{U,V,W} : \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee) // G \dashrightarrow |\mathcal{O}_{\mathbb{P}U}(3)| // \text{SL}(U).$$

The main results of this paper are the following:

Theorem A. *A point $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ is G -stable if and only if the associated cubic surface $\det(\varphi) \in |\mathcal{O}_{\mathbb{P}U}(3)|$ is $\text{SL}(U)$ -stable. The point φ is G -semistable if and only if either:*

- *the cubic surface $\det(\varphi)$ is $\text{SL}(U)$ -semistable; or*

- the cubic surface $\det(\varphi)$ has either one A_3 singularity or both one A_1 and one A_3 singularity (and is otherwise smooth) and the associated subscheme $X_\varphi \subset \mathbb{P}V$ is supported at only two points; or
- the cubic surface $\det(\varphi)$ is reducible and the associated subscheme $X_\varphi \subset \mathbb{P}V$ is a smooth conic.

The quotient $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)^{G-ss}/G$ of the strictly G -semistable locus is the disjoint union of a singleton and a subscheme whose normalisation is $\mathbb{P}^1$. The rational map $\det_{UVW}$ is undefined only at one point.

1.1. A brief history. Much of the theory of determinantal representations of normal cubic surfaces was known by classical geometers such as Segre, who proved in 1906 that every normal cubic surface without an E_6 singularity has a determinantal representation [Seg06]. Thrall then published a classification of $4 \times 3 \times 3$ tensors in 1941 [Thr41]. This was supplemented by Ng in 2002 [Ng02], who wrote down an explicit matrix presentation for almost all $4 \times 3 \times 3$ tensors.

In 1989, Gimigliano introduced a sheaf-theoretic approach to smooth determinantal surfaces [Gim89] by viewing the various adjoint matrices as maps of vector bundles on $\mathbb{P}^2$ and $\mathbb{P}^3$. This was extended by Dolgachev and Kapranov in [DK93], who described the geometry of smooth determinantal cubic surfaces. Dolgachev gives a detailed description of an extension of this sheaf-theoretic approach to cubic surfaces with at worst canonical singularities in [Dol12, §9.3]. Eisenbud and Popescu's two papers on the Gale transform in the late 90s [EP99; EP00] give a modern treatment of the connections between the Gale transform, free resolutions of ideals of points and determinantal varieties. The special case of $4 \times 3 \times 3$ tensors is a very well-behaved example of this connection.

The connection between determinantal representations of cubic surfaces and twisted cubics was studied by Ellingsrud, Pienie and Strømme in 1987 [EPS87] and then more recently by Lehn, Lehn, Sorger and Straten in [LLSS17]. The latter paper uses the geometry of the GIT quotient $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)/(\mathrm{SL}(V) \times \mathrm{SL}(W))$ to describe the moduli space of twisted cubics on a smooth cubic fivefold.

1.2. Structure of the paper. Section 2 introduces important notation and recounts some known facts about determinantal representations. Most of these results, with more details and proofs, can be found in either [Dol12, §9] or [Ng02] or [LLSS17].

Section 3 uses Gale-duality to construct the birational contraction

$$\mathrm{Hilb}_6(\mathbb{P}^2) \dashrightarrow \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)/(\mathrm{SL}(V) \times \mathrm{SL}(W)),$$

and we describe where this map is defined.

Section 4 is devoted to understanding the full GIT quotient by $\mathrm{SL}(U) \times \mathrm{SL}(V) \times \mathrm{SL}(W)$ and proving Theorem A. We give a stratification of $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$, which allows us to describe the G -semistable and G -stable loci, and the identification of G -semistable orbits in the GIT quotient.

1.3. Conventions and organisation. We are working over the complex numbers. We use the subspace convention for the projective space $\mathbb{P}A$ associated to a vector space A , i.e. $\mathbb{P}A$ is the projective space of 1-dimensional subspaces of A ; and $\mathbb{P}A = \mathrm{Proj}(\mathbb{C}[A^\vee]_\bullet)$ and $H^0(\mathbb{P}A, \mathcal{O}_{\mathbb{P}A}(1)) = A^\vee$. We use the same convention for the projectivisation of a vector bundle and do not distinguish between a vector bundle and its associated sheaf of sections. When A is a vector space we will always let $\mathrm{SL}(A)$ act on A with the standard left action $a \mapsto g \cdot a$ and on $A^\vee$ by the left action $\alpha \mapsto \alpha \circ g^{-1}$ of precomposition. For convenience of notation, we will not distinguish between non-zero vectors $a \in A$ and their projective equivalence class $\bar{a} \in \mathbb{P}A$.

All schemes are of finite type over $\mathbb{C}$. Unless otherwise specified, all tensor products are taken either over $\mathbb{C}$ or over the structure sheaf of the ambient scheme. All *points* are $\mathbb{C}$ -points unless otherwise specified and by a *finite scheme* we mean a scheme with finite structure morphism to $\text{Spec } \mathbb{C}$. When X is a finite scheme, the *length* of a point $x \in X$ is the length of the non-reduced subscheme supported at x . When $X \subset Y$ is a closed subscheme we will write $\mathcal{I}_X \subset \mathcal{O}_Y$ for the ideal sheaf that cuts out $X \subset Y$. When $\mathcal{F}$ is a coherent sheaf on a scheme X we will write $|\mathcal{F}| = \mathbb{P}(H^0(X, \mathcal{F}))$ for the complete linear system.

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2. BACKGROUND

Recall that U and V and W are vector spaces with $\dim U = 4$ and $\dim V = \dim W = 3$. The vector spaces V and W play completely symmetric roles in this paper. Whenever we fix bases of $U^\vee$ and $V^\vee$ and $W^\vee$ we will write the basis vectors as u_0, u_1, u_2, u_3 and v_0, v_1, v_2 and w_0, w_1, w_2 .

As outlined in the introduction, a general tensor $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ is adjoint to (a) a matrix $\varphi_U \in U^\vee \otimes \text{Hom}(W, V^\vee)$ whose determinant $\det(\varphi_U)$ cuts out a smooth cubic surface $S_\varphi \subset \mathbb{P}U$ and (b) the presentation matrix φ_V of a free resolution of an ideal sheaf $\mathcal{I}_{X_\varphi} \subset \mathcal{O}_{\mathbb{P}V}$ that cuts out a length 6 subscheme $X_\varphi \subset \mathbb{P}V$. The cubic surface S_φ is isomorphic to the blowup of $\mathbb{P}V$ at X_φ , the

Definition 2.3. We call F_S the *associated Dynkin diagram* to the surface S . We say a cubic surface *has F singularities* if its associated Dynkin diagram is isomorphic to F (as a graph). We say a cubic surface S *has at worst F singularities* if its associated Dynkin diagram is isomorphic (as a graph) to a Dynkin subdiagram of F .

For each strict Dynkin subdiagram $F \subsetneq \tilde{E}_6$ we let $\mathcal{S}(F) \subset |\mathcal{O}_{\mathbb{P}U}(3)|$ denote the set of cubic surfaces $S \subset \mathbb{P}U$ with F singularities.

The subsets $\mathcal{S}(F) \subset |\mathcal{O}_{\mathbb{P}U}(3)|$ are connected, locally closed subvarieties and $\mathcal{S}(F') \subset \overline{\mathcal{S}(F)}$ if and only if F is a Dynkin subdiagram of F' .

Proposition 2.4. *The subvariety $\mathcal{S}(\emptyset) \subset |\mathcal{O}_{\mathbb{P}U}(3)|$ parametrising smooth cubic surfaces is dense in $|\mathcal{O}_{\mathbb{P}U}(3)|$ and every smooth cubic surface has a finite automorphism group ($\mathrm{PGL}(U)$ -stabiliser). More generally:*

- *Let $F \subsetneq \tilde{E}_6$ be a Dynkin subdiagram where $F \neq 2A_2$ and $F \neq D_4$. Then $\mathcal{S}(F) \subset |\mathcal{O}_{\mathbb{P}U}(3)|$ has codimension $|F|$ and every surface $[S] \in \mathcal{S}(F)$ has a $\min\{0, |F| - 4\}$ -dimensional automorphism group. If additionally $|F| \geq 4$ then all surfaces with F singularities are isomorphic and $\mathcal{S}(F)$ is a $\mathrm{PGL}(U)$ -orbit.*
- *Let $F = 2A_2$. Then the subvariety $\mathcal{S}(2A_2) \subset |\mathcal{O}_{\mathbb{P}U}(3)|$ has codimension 3 and every surface $[S] \in \mathcal{S}(2A_2)$ has a 1-dimensional automorphism group.*
- *Let $F = D_4$. Then the subvariety $\mathcal{S}(D_4) \subset |\mathcal{O}_{\mathbb{P}U}(3)|$ has codimension 4. There are two types of D_4 cubic surfaces up to isomorphism, we call them $S_{D_4(a)}$ and $S_{D_4(b)}$. We have $\mathrm{Aut}(S_{D_4(a)}) = \mathfrak{S}_3$ and $\mathrm{Aut}(S_{D_4(b)}) = \mathbb{G}_m \ltimes \mathfrak{S}_3$. The orbit $\mathrm{PGL}(U) \cdot [S_{D_4(a)}] \subset \mathcal{S}(D_4)$ is dense.*

The automorphism groups of cubic surfaces with F singularities satisfying both $|F| \geq 4$ and $F \neq 2A_2$ can be found in [Sak10]. Section 8 of [CP21] lists all positive-dimensional automorphism groups of canonical cubic surfaces.

2.2. Point configurations in $\mathbb{P}^2$. Let $X \subset \mathbb{P}V \cong \mathbb{P}^2$ be a configuration of 6 points in general position with respect to plane curves (no 3 points in X are collinear and X is not contained in a conic). The blowup $b : S = \mathrm{Bl}_X \mathbb{P}V \rightarrow \mathbb{P}V$ is a smooth cubic surface with anticanonical divisor $\omega_S^{-1} = b^* \mathcal{I}_{\mathbb{P}V}(3)$. Conversely, let S' be a smooth cubic surface. Every birational morphism $S' \rightarrow \mathbb{P}V$ is the blowup of $\mathbb{P}V$ at 6 points $X \subset \mathbb{P}V$ which are in general position with respect to plane curves. There are exactly 72 birational morphisms $S' \rightarrow \mathbb{P}V$, up to $\mathrm{PGL}(V)$ -equivalence on the target.

Canonical cubic surfaces are similarly described by certain curvilinear length 6 subschemes of $\mathbb{P}V$. We introduce some notation, closely following Subsection 2.5 of [DPT21] (note that our definition of canonical is their definition of admissible). We are also interested in length 6 subschemes $Y \subset \mathbb{P}W \cong \mathbb{P}^2$. We will mainly write definitions and theorems in terms of $X \subset \mathbb{P}V$, in both the following definition and the rest of the paper the reader should replace all V s with W s and all X s with Y s to apply the appropriate statements to subschemes of $\mathbb{P}W$; the two 3-dimensional vector spaces V and W play completely symmetric roles.

Definition 2.5. A length 6 subscheme $X \subset \mathbb{P}V$ is said to be *canonical* the following conditions are all satisfied:

- (i) X is curvilinear; and
- (ii) $h^0(\mathbb{P}V, \mathcal{I}_X(3)) = 4$; and
- (iii) the twisted ideal sheaf $\mathcal{I}_X(3)$ is generated by its global sections.

A canonical subscheme $X \subset \mathbb{P}V$ is said to be *admissible* if additionally:

(iv) X is not contained in a conic.

An admissible subscheme $X \subset \mathbb{P}V$ is said to be a *smooth point configuration* if additionally:

- (v) X is reduced; and
- (vi) X contains no length 3 collinear subscheme.

Smooth point configurations $X \subset \mathbb{P}V$ are simply sets of 6 points in general position with respect to plane curves, and are associated to smooth cubic surfaces. For canonical subschemes we have the following result:

Proposition 2.6. *Let $X \subset \mathbb{P}V$ be a canonical length 6 subscheme. Let $S = \text{Bl}_X \mathbb{P}V$ and let $b : S = \text{Bl}_X \mathbb{P}V \rightarrow \mathbb{P}V$ be the blowup. The following facts are true:*

- (i) S has at worst A_n singularities.
- (ii) The linear system $H^0(S, \omega_S^{-1})$ is 4-dimensional and basepoint-free.
- (iii) The image of the map $f : S \rightarrow \mathbb{P}U \cong |\omega_S^{-1}|$ induced by $\omega_S^{-1} = b^* \mathcal{I}_X(3)$ is a cubic surface $S_X \subset \mathbb{P}U$, the morphism $S \rightarrow S_X$ is birational.
- (iv) The morphism $S \rightarrow S_X$ is an isomorphism away from the following curves, which it contracts:
 - the strict transform $b^*(C)$ of any conic $C \subset \mathbb{P}V$ containing X , and
 - the strict transform $b^*(L)$ of any line $L \subset \mathbb{P}V$ such that $\text{len}(X \cap L) = 3$.
- (v) S_X is normal and has only canonical singularities.

Conversely, let S' be a canonical cubic surface and let $\tilde{S}' \rightarrow S'$ be the minimal resolution of singularities. Any birational morphism $\tilde{S}' \rightarrow \mathbb{P}V$ factors through the blowup $\text{Bl}_X \mathbb{P}V \rightarrow \mathbb{P}V$ of $\mathbb{P}V$ at a canonical subscheme $X \subset \mathbb{P}V$. The surface S' is isomorphic to S_X as above.

Proof. The details of (i-iii) and (v) can be found in Proposition 2.5.2 of [DPT21]. We have added (iv), which is easy to verify using basic facts about linear systems. The case when S_X is smooth can be found in detail in Proposition IV.9 of [Bea78].

The details of the converse statement can be found in Chapter 9 of [Dol12], which uses the language of bubble cycles instead of length 6 subschemes. $\square$

The relationship between a canonical subscheme $X \subset \mathbb{P}V$, the blowup $b : \text{Bl}_X \mathbb{P}V \rightarrow \mathbb{P}V$, the associated cubic surface S_X , and the minimal resolution of singularities $r : \tilde{S} \rightarrow S_X$ can be summarised in the following commutative diagram where all morphisms are birational:

$$(3) \quad \begin{array}{ccccc} & & \tilde{S} & & \\ & \swarrow & \downarrow r & \searrow & \\ \text{Bl}_X \mathbb{P}V = S & \xrightarrow{f} & S_X & \hookrightarrow & |\omega_S^{-1}| \cong \mathbb{P}^3 \\ & \searrow b & \downarrow \text{---} & & \\ & & \mathbb{P}V & & \end{array}$$

The birational map $f^{-1} \circ r : \tilde{S} \dashrightarrow \text{Bl}_X \mathbb{P}V$ extends to a morphism; this morphism is the minimal resolution of singularities for the blowup $\text{Bl}_X \mathbb{P}V$. The composition $b \circ f^{-1} \circ r : \tilde{S} \rightarrow \mathbb{P}V$ is the composition of 6 blowups of $\mathbb{P}V$ (blowing up the bubble cycle associated to the subscheme $X \subset \mathbb{P}V$). The birational map $f \circ b^{-1} : \mathbb{P}V \dashrightarrow S_X$ is given by the linear system $|\mathcal{I}_X(3)|$ of cubics through X . If $X \subset \mathbb{P}V$ is also admissible (i.e. not contained in a conic) then the pullback $(b \circ f^{-1})^* \mathcal{O}_{\mathbb{P}V}(1)$ under the birational map $b \circ f^{-1} : S_X \dashrightarrow \mathbb{P}V$ is a line bundle on S_X associated to a twisted cubic.

Definition 2.7. Let $X \subset \mathbb{P}V$ be a canonical subscheme. We define *the cubic surface associated to X* , denoted by S_X , as the image of the map $\mathrm{Bl}_X \mathbb{P}V \rightarrow \mathbb{P}U$ induced by the linear system $|\omega_{\mathrm{Bl}_X \mathbb{P}V}^{-1}|$.

Conversely, if S is a cubic surface and $X \subset \mathbb{P}V$ is a canonical subscheme where $S \cong S_X$ then X is said to be *an associated subscheme* to the surface S .

The associated cubic surface S_X to a canonical subscheme $X \subset \mathbb{P}V$ is always unique up to isomorphism, however in general there are multiple (at most 72) non-isomorphic canonical subschemes associated to a given canonical cubic surface.

Remark 2.8. The construction sending a canonical subscheme X to its associated cubic surface S_X can also be done in families. For ease of notation we only write it over affine schemes. Let $\mathrm{Spec} A$ be an affine scheme and let $X_A \subset \mathbb{P}V_A := \mathbb{P}V \times \mathrm{Spec} A$ be a closed subscheme, flat over $\mathrm{Spec} A$, whose fibres are canonical length 6 subschemes. The blowup $\mathrm{Bl}_X \mathbb{P}V_A$ is A -flat and the associated family of cubic surfaces

$$\mathcal{S}_{X_A} = \mathrm{Proj}_A \left(\bigoplus_{n \geq 0} H^0(\mathrm{Bl}_X \mathbb{P}V_A, \omega_{\mathrm{Bl}_X \mathbb{P}V_A}^{-n}) \right)$$

is A -flat (see Proposition 2.5.4 of [DPT21]). The family $\mathcal{S}_{X_A}$ is embedded in the projectivisation of the rank 4, locally-free A -module $H^0(\mathrm{Bl}_{X_A} \mathbb{P}V_A, \omega_{\mathrm{Bl}_{X_A} \mathbb{P}V_A}^{-1})$. Note that under the blowup $f : \mathrm{Bl}_{X_A} \mathbb{P}V_A \rightarrow \mathbb{P}V_A$ we have $f^* \mathcal{I}_{X_A}(3) = \omega_{\mathrm{Bl}_{X_A} \mathbb{P}V_A}^{-1}$. This induces an isomorphism of locally-free A -modules

$$f^* : H^0(\mathbb{P}V_A, \mathcal{I}_X(3)) \longrightarrow H^0(\mathrm{Bl}_X \mathbb{P}V_A, \omega_{\mathrm{Bl}_{X_A} \mathbb{P}V_A}^{-1})$$

because the map $f^* \otimes \mathrm{id}_{A/\mathfrak{a}} : H^0(\mathbb{P}V, \mathcal{I}_{X_{\mathfrak{a}}}(3)) \rightarrow H^0((\mathrm{Bl}_X \mathbb{P}V_A)_{\mathfrak{a}}, \omega_{(\mathrm{Bl}_{X_A} \mathbb{P}V_A)_{\mathfrak{a}}}^{-1})$ is an isomorphism for every maximal ideal $\mathfrak{a} \subset A$.

The flat family $\mathcal{S}_{X_A} \rightarrow \mathrm{Spec} A$ induces a rational map

$$\gamma : \mathrm{Hilb}_6(\mathbb{P}V) \dashrightarrow |\mathcal{O}_{\mathbb{P}U}(3)| // \mathrm{SL}(U),$$

which is defined on the locus of canonical subschemes X whose associated cubic surface S_X is $\mathrm{SL}(U)$ -semistable. A result of Mumford [Mum77, Subsection 1.14] states that a surface S is $\mathrm{SL}(U)$ -semistable if and only if S has at most $4A_1$ or $3A_2$ singularities (equivalently, if all of the singularities on S are either of type A_1 or A_2).

The set of singular cubic surfaces $\Delta \subset |\mathcal{O}_{\mathbb{P}U}(3)|$ forms an irreducible divisor of degree 32 [Sal60; Cle61b] and the subvariety $\mathcal{S}(A_1) \subset \Delta$ is dense. However, the set of length 6 subschemes associated to a singular cubic surface does not form (an open subset of) an irreducible divisor inside $\mathrm{Hilb}_6(\mathbb{P}V)$. Concretely, a general (automorphism-free) A_1 cubic surface S has 51 associated subschemes $X \subset \mathbb{P}V$ (up to $\mathrm{PGL}(3)$ -equivalence). These subschemes are split into the following three types:

- (i) 30 subschemes which contain a length 2 point; and
- (ii) 20 subschemes which contain a length 3 collinear subscheme; and
- (iii) 1 subscheme contained in a conic.

Other than satisfying exactly one of the above three conditions, a subscheme $X \subset \mathbb{P}V$ associated to an A_1 cubic surface is otherwise reduced and in general position with respect to plane curves.

A consequence of the Hilbert-Burch theorem (see Subchapter 20.4 of [Eis95]) is that the ideal sheaves $\mathcal{I}_X \subset \mathcal{O}_{\mathbb{P}V}$ cutting out a subscheme $X \subset \mathbb{P}V$ of types (i) and (ii) have a free resolution of the same shape, but the ideal sheaf cutting out a subscheme of type (iii) does not. More precisely,

Proposition 2.9. *The ideal sheaf $\mathcal{I}_X$ cutting out a canonical length 6 subscheme $X \subset \mathbb{P}V$ has a free resolution of the form*

$$(4) \quad 0 \longrightarrow \mathcal{O}_{\mathbb{P}V}^{\oplus 3}(-4) \longrightarrow \mathcal{O}_{\mathbb{P}V}^{\oplus 4}(-3) \longrightarrow \mathcal{I}_X \longrightarrow 0$$

if and only if X is admissible (not contained in a conic). When such a free resolution exists, it is unique up to the $\mathrm{GL}(4) \times \mathrm{GL}(3)$ action on $\mathrm{Hom}_{\mathcal{O}_{\mathbb{P}V}}(\mathcal{O}_{\mathbb{P}V}^{\oplus 3}(-4), \mathcal{O}_{\mathbb{P}V}^{\oplus 4}(-3)) \cong \mathrm{Hom}(\mathbb{C}^3, \mathbb{C}^4)$.

Proof. A proof can be found in Subchapter 9.3.2 of [Dol12], a proof for the case when X is reduced is also found in Subchapter 20.4.1 of [Eis95]. We only give a sketch of Dolgachev's proof.

Let $X \subset \mathbb{P}V$ be an admissible subscheme cut out by an ideal $I_X \subset \mathbb{C}[V^\vee]_\bullet$. The ideal I_X is generated by 4 linearly independent cubic polynomials and has $\mathrm{depth}(I_X) = \mathrm{codim}(I_X) = 2$, so I_X satisfies the conditions of the Hilbert-Burch theorem. Then there is a 4×3 matrix φ_1 valued in $\mathbb{C}[V^\vee]_\bullet$ and a polynomial $a \in \mathbb{C}[V^\vee]_\bullet$ such that $I_X = (am_0, \dots, am_3)$, where $m_0, \dots, m_3$ are the maximal 3×3 minors of φ_1 . Then a is a unit and $\varphi_1 \in V^\vee \otimes \mathrm{Hom}(\mathbb{C}^3, \mathbb{C}^4)$ is valued in linear forms in V , and $I_X = (m_0, \dots, m_3)$. The matrix φ_1 presents a free resolution

$$0 \longrightarrow \mathbb{C}[V^\vee]_\bullet^{\oplus 3}(-4) \xrightarrow{\varphi_1} \mathbb{C}[V^\vee]_\bullet^{\oplus 4}(-3) \longrightarrow I_X \longrightarrow 0$$

which, after taking the projective spectrum, gives the exact sequence (4).

The matrix presenting any free resolution of the form of (4) is uniquely determined by the $V^\vee$ -linear relations between the cubic generators of I_X . These relations are the rows of the matrix φ_1 , unique up to the $\mathrm{GL}(4)$ -freedom of the choice of basis for the set of generators of I_X and a $\mathrm{GL}(3)$ -freedom of the choice of basis for the set of relations.

Conversely, if X is contained in a conic $\{f_2 = 0\}$ then X is the complete intersection of the conic $\{f_2 = 0\}$ with a cubic $\{f_3 = 0\}$. The ideal $I_X \subset \mathbb{C}[V^\vee]_\bullet$ is generated by the polynomials f_2 and f_3 . The Hilbert-Burch theorem still gives a free resolution of I_X as

$$0 \longrightarrow \mathbb{C}[V^\vee]_\bullet(-5) \xrightarrow{\varphi_1} \mathbb{C}[V^\vee]_\bullet(-2) \oplus \mathbb{C}[V^\vee]_\bullet(-3) \longrightarrow I_X \longrightarrow 0,$$

presented by the matrix $\varphi_1 = \begin{pmatrix} -f_3 & f_2 \end{pmatrix}$, and I_X is generated by the two maximal minors of φ_1 . But this is not a free resolution given by a 4×3 matrix of linear forms and any attempt to construct such a resolution fails: the matrix of $V^\vee$ -linear relations between the cubic generators $v_0 f_2, v_1 f_2, v_2 f_2, f_3$ is

$$\begin{pmatrix} 0 & v_2 & -v_1 \\ -v_2 & 0 & v_0 \\ v_1 & -v_0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

whose 3×3 minors all vanish. □

After choosing identifications $U^\vee \cong \mathbb{C}^4$ and $W \cong \mathbb{C}^3$, the matrix $\varphi_1 \in V^\vee \otimes \mathrm{Hom}(\mathbb{C}^3, \mathbb{C}^4)$ is adjoint to a tensor $\varphi \in U^\vee \otimes V^\vee \otimes W^\vee$. The GIT moduli space of tensors $\varphi \in U^\vee \otimes V^\vee \otimes W^\vee$ is the focus of this paper.

2.3. Determinantal representations.

Definition 2.10. Let $\varphi \in U^\vee \otimes V^\vee \otimes W^\vee$ be a $4 \times 3 \times 3$ tensor. The tensor φ has several adjoints we are interested in; for each decomposition of $U \otimes V \otimes W$ as $A \otimes B$ we will write φ_A for the adjoint linear map

$$\varphi_A : A \longrightarrow B^\vee.$$

For example, $\varphi_U \in \mathrm{Hom}(U, V^\vee \otimes W^\vee)$ and $\varphi_{U \otimes W} \in \mathrm{Hom}(U \otimes W, V^\vee)$. When $A, B, C \in \{U, V, W\}$ are three distinct vector spaces we will often view $\varphi_A : A \rightarrow B^\vee \otimes C^\vee = \mathrm{Hom}(B, C^\vee)$ as an

element of $A^\vee \otimes \text{Hom}(B, C^\vee)$ which, after choosing bases for B and C , can be identified with a $\dim(C) \times \dim(B)$ matrix of linear forms in A . As a convention, we will view φ_U as an element of $U^\vee \otimes \text{Hom}(W, V^\vee)$; the vector space $W^\vee$ indexes the rows and $V^\vee$ indexes the columns.

Definition 2.11. Let $\varphi_A \in \text{Hom}(A, B^\vee \otimes C^\vee)$ be a tensor. For each $r \leq \dim A$ we let the *rank at most r subscheme* $\text{rk}_{\leq r}(\varphi_A) \subset \mathbb{P}A$ be the subscheme of points $a \in A$ where $\varphi_A(a)$ has rank at most r . We say a point $a \in \text{rk}_{\leq r}(\varphi_A)$ is a *rank at most r point* of φ_A .

The rank at most 2 subschemes of φ will be given special names.

- We let $S_\varphi := \text{rk}_{\leq 2}(\varphi_U) \subset \mathbb{P}U$, which is cut out by the determinant cubic $\det(\varphi_U)$. We call the scheme S_φ the *associated cubic surface* to φ .
- We let $X_\varphi := \text{rk}_{\leq 2}(\varphi_V) \subset \mathbb{P}V$.
- We let $Y_\varphi := \text{rk}_{\leq 2}(\varphi_W) \subset \mathbb{P}W$. The schemes X_φ and Y_φ are cut out by the maximal 3×3 minors of φ_V and φ_W , respectively. We call X_φ and Y_φ the *associated point configurations* to φ .

In general S_φ is a smooth cubic surface, the subschemes X_φ and Y_φ are smooth point configurations, and the rank at most 1 subschemes $\text{rk}_{\leq 1}(\varphi_U)$ and $\text{rk}_{\leq 1}(\varphi_V)$ and $\text{rk}_{\leq 1}(\varphi_W)$ are empty. The derivative of the determinant $\det(\varphi_U)$ vanishes if all the 2×2 minors vanish, so every point of $\text{rk}_{\leq 1}(\varphi_U)$ is a singularity of S_φ . The converse is not true.

Definition 2.12. Let $S \subset \mathbb{P}U$ be a cubic surface (resp. $X \subset \mathbb{P}V$, resp. $Y \subset \mathbb{P}W$ be subschemes). A *determinantal representation* of S (resp. X , resp. Y) is a tensor $\varphi \in U^\vee \otimes V^\vee \otimes W^\vee$ where $S = S_\varphi$ (resp. $X = X_\varphi$, resp. $Y = Y_\varphi$).

Every cubic surface $S \subset \mathbb{P}U$ has a determinantal representation except for the E_6 cubic surface (whose defining polynomial is projectively equivalent to $u_0^2 u_3 + u_0 u_2^2 + u_1^3$). This was known to Segre in [Seg06], a proof can be found in Theorem 7.6 of [Thr41]. When S is canonical the structure (and existence) of these determinantal representations is completely characterised by its set of associated subschemes, following the construction outlined in Proposition 2.9.

Proposition 2.13. Fix a tensor $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$. The following are equivalent:

- (i) S_φ is canonical.
- (ii) X_φ is admissible.
- (iii) Y_φ is admissible.

If S_φ is canonical then S_φ is the associated cubic surface to X_φ and Y_φ .

Every admissible length 6 subscheme of $\mathbb{P}V$ has a unique determinantal representation, up to $\text{GL}(U) \times \text{GL}(W)$ -equivalence. The set of determinantal representations of a canonical cubic surface S is in bijection (up to $\text{GL}(U) \times \text{GL}(V) \times \text{GL}(W)$ -equivalence) with the set of admissible length 6 subschemes $X \subset \mathbb{P}V$ whose associated cubic surface is S .

Proof. All details can be found in [Dol12, §9.3.2]. □

2.4. Low rank linear subspaces. A useful $\text{GL}(U) \times \text{GL}(V) \times \text{GL}(W)$ -invariant of a tensor $\varphi \in U^\vee \otimes V^\vee \otimes W^\vee$ is the number and type of linear subspaces on which φ drops rank (linear subspaces of $\text{rk}_{\leq r}(\varphi_A)$). We will use these linear subspaces for our GIT semistability and stability classification. We introduce some notation and facts about these subspaces, all proofs and details can be found in the papers [AL80] and [AL81] by Atkinson and Lloyd.

Definition 2.14. Let A and B be vector spaces. A subspace $C \subset A^\vee \otimes B^\vee$ is said to be of *block type* (a, b) if there exist subspaces $A' \subset A$ and $B' \subset B$ with $\dim A' = a$ and $\dim B' = b$ such that $A' \otimes B' \subset \ker(c : A \otimes B \rightarrow \mathbb{C})$ for every pairing $c \in C$.

Let r be an integer. A subspace $C \subset A^\vee \otimes B^\vee$ is said to be a *rank at most r subspace* if every pairing $c \in C$ has rank at most r .

Up to a choice of basis for A and B , each subspace of block type (a, b) is of the form

$$\left[\begin{array}{c|c} * & * \\ \hline * & 0 \end{array} \right] \subset A^\vee \otimes B^\vee$$

with a $a \times b$ block of zeros in the corner.

It is easy to verify that every subspace of block type (a, b) is a rank at most r subspace where $r = \dim A + \dim B - a - b$, but the converse is not true. The classification of rank at most r subspaces is in general quite difficult, but we only need the cases of $r = 1$ and $r = 2$.

Proposition 2.15 ([AL81, Lemma 2]). *Every rank 1 subspace satisfies a rank 1 block condition.*

For rank at most 2 subspaces we introduce some notation to describe the only exception, which is the vector space of skew-symmetric 3×3 matrices.

Definition 2.16. Let A and B be vector spaces and let $C \subset A^\vee \otimes B^\vee$ be a 3-dimensional subspace. If there exist injections $\iota_A : \mathbb{C}^3 \rightarrow A^\vee$ and $\iota_B : \mathbb{C}^3 \rightarrow B^\vee$ such that

$$C = (\iota_A \otimes \iota_B)(\mathbb{C}^3 \wedge \mathbb{C}^3) \subset A^\vee \otimes B^\vee$$

then C is said to be a *skew-symmetric subspace* of $A^\vee \otimes B^\vee$.

An interesting property of skew-symmetric subspaces is the following:

Proposition 2.17. *Let A and B and C be 3-dimensional vector spaces. Let $\varphi \in A^\vee \otimes B^\vee \otimes C^\vee$ be a $3 \times 3 \times 3$ tensor. All three of the subspaces $\varphi_A(A) \subset B^\vee \otimes C^\vee$ and $\varphi_B(B) \subset A^\vee \otimes C^\vee$ and $\varphi_C(C) \subset A^\vee \otimes B^\vee$ are skew-symmetric subspaces if and only if just one of them is a skew-symmetric subspace.*

Proof. Suppose $\varphi_C(C) \subset A^\vee \otimes B^\vee$ is a skew-symmetric subspace. Then there is a choice of basis for A and B and C that identifies $\varphi_C : C \rightarrow A^\vee \otimes B^\vee$ with the Hodge star map $\star : \mathbb{C}^3 \rightarrow \wedge^2(\mathbb{C}^3)^\vee$. The Hodge star map $\star$ is adjoint to a non-zero element of $\wedge^3(\mathbb{C}^3)^\vee \cong \mathbb{C}$, the proposition follows by the symmetry of the three adjoints of $\star$. $\square$

Proposition 2.18 ([AL81, Lemma 7]). *Every rank at most 2 subspace either satisfies a rank 2 block condition or is a (3-dimensional) skew-symmetric subspace.*

The class of tensors $\varphi \in U^\vee \otimes V^\vee \otimes W^\vee$ whose associated cubic surface S_φ contains a skew-symmetric subspace will appear repeatedly throughout this paper.

Definition 2.19. A tensor $\varphi \in U^\vee \otimes V^\vee \otimes W^\vee$ is said to be *skew-symmetric* if there is a 3-dimensional linear subspace $U' \subset U$ such that $\varphi_U(U') \subset V^\vee \otimes W^\vee$ is a skew-symmetric subspace.

Up to $\mathrm{GL}(U) \times \mathrm{GL}(V) \times \mathrm{GL}(W)$ -equivalence, a skew-symmetric tensor φ can be written as

$$\varphi_U = u_0 Q + \begin{pmatrix} 0 & u_3 & -u_2 \\ -u_3 & 0 & u_1 \\ u_2 & -u_1 & 0 \end{pmatrix} \in U^\vee \otimes \mathrm{Hom}(W, V^\vee),$$

where $U' = \ker(u_0) \subset U$. We define the *rank* of the skew-symmetric tensor φ to be the rank of the matrix Q . We let $\Sigma \subset \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ be the subvariety of skew-symmetric tensors. We let $\Sigma_r \subset \Sigma$ be the subvariety of skew-symmetric tensors $\varphi \in \Sigma$ of rank r .

When $\varphi \in U^\vee \otimes V^\vee \otimes W^\vee$ is a skew-symmetric tensor, the associated 3-dimensional subspace $U' \subset U$ is unique. The fact that $\varphi_U(U') \subset V^\vee \otimes W^\vee$ is a skew-symmetric subspace is equivalent to the following commutative diagram being induced by three isomorphisms $U' \cong V \cong W \cong \mathbb{C}^3$:

$$\begin{array}{ccccc} U' & \xrightarrow{\varphi_U|_{U'}} & V^\vee \otimes W^\vee & & \\ \sim \downarrow & & \downarrow & \searrow \sim & \\ \mathbb{C}^3 & \xrightarrow{\star} & \wedge^2(\mathbb{C}^3)^\vee & \hookrightarrow & (\mathbb{C}^3)^\vee \otimes (\mathbb{C}^3)^\vee. \end{array}$$

As a result, a skew-symmetric tensor φ_U fixes an identification $U' \cong V \cong W$.

Up to $\mathrm{GL}(U)$ -equivalence, the matrix $Q \in \mathrm{Hom}(W, V^\vee)$ can be chosen to be symmetric with respect to this identification $V \cong W$. There is a symmetric matrix $B \in \mathrm{GL}(V) \cong \mathrm{GL}(W)$ that diagonalises A and preserves the skew-symmetric component of φ_U (the matrix B also acts on U' , preserving the isomorphism $U' \cong V \cong W$). The eigenvalues of $B^T Q B$ can be normalised to either 0 or 1 (depending on the rank of Q) and can be permuted by permuting the bases of U and V and W . As a result,

Proposition 2.20. *Every skew-symmetric tensor with the same rank is $\mathrm{GL}(U) \times \mathrm{GL}(V) \times \mathrm{GL}(W)$ -equivalent.*

The rank at most 2 subschemes associated to a skew-symmetric tensor $\varphi \in \Sigma_r$ are easy to compute from the 3×3 minors of the matrices φ_U and φ_V and φ_W . These are:

- Suppose $\varphi \in \Sigma_3$. The surface $S_\varphi \subset \mathbb{P}U$ is the transverse union of a (skew-symmetric) plane and a smooth quadric surface. The subschemes $X_\varphi \subset \mathbb{P}V$ and $Y_\varphi \subset \mathbb{P}W$ are smooth conics.
- Suppose $\varphi \in \Sigma_2$. The surface $S_\varphi \subset \mathbb{P}U$ is the transverse union of three planes (one skew-symmetric and two of type $(2, 2)$). The subschemes $X_\varphi \subset \mathbb{P}V$ and $Y_\varphi \subset \mathbb{P}W$ are the transverse unions of two lines.
- Suppose $\varphi \in \Sigma_1$. The surface $S_\varphi \subset \mathbb{P}U$ is the transverse union of a (skew-symmetric) plane and a doubled plane (of type $(2, 2)$). The subschemes $X_\varphi \subset \mathbb{P}V$ and $Y_\varphi \subset \mathbb{P}W$ are doubled lines.
- Suppose $\varphi \in \Sigma_0$. Then $S_\varphi = \mathbb{P}U$ and $X_\varphi = \mathbb{P}V$ and $Y_\varphi = \mathbb{P}W$.

3. PARTIAL GIT QUOTIENTS

Before looking at the full Geometric Invariant Theory (GIT) quotient of $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ by $\mathrm{SL}(U) \times \mathrm{SL}(V) \times \mathrm{SL}(W)$ we will review the partial quotients by the subgroups $\mathrm{SL}(U)$ and $\mathrm{SL}(V)$ and $\mathrm{SL}(W)$, and $\mathrm{SL}(U) \times \mathrm{SL}(V)$ and $\mathrm{SL}(U) \times \mathrm{SL}(W)$ and $\mathrm{SL}(V) \times \mathrm{SL}(W)$. These are all special cases of quiver GIT. We refer the reader to either [Hos16] or [MFK94] for details on GIT.

Let H be a reductive group acting linearly on a vector space T . A closed point $t \in \mathbb{P}T$ is said to be *H-semistable* if there exists a non-constant H -invariant function $f \in \mathbb{C}[T^\vee]_\bullet$ where $f(t) \neq 0$. The point t is said to be *H-stable* if additionally the stabiliser $\mathrm{Stab}_H(t)$ is finite and the H -orbit $H \cdot t$ is closed inside the set of H -semistable points.

Let $Z \subset \mathbb{P}T$ be a H -invariant subscheme. We will write

$$Z^{H-s} := \{z \in Z \mid z \text{ is } H\text{-stable}\}$$

for the *H-stable* locus,

$$Z^{H-ss} := \{z \in Z \mid z \text{ is } H\text{-semistable}\}$$

for the *H-semistable* locus, and we will write $Z^{H-sss} := Z^{H-ss} \setminus Z^{H-s}$ for the *strictly H-semistable* locus. Note that $Z^{H-s} \subset Z^{H-ss}$ and $Z^{H-ss} \subset Z$ are (possibly empty) open subschemes. A point

$t \in \mathbb{P}T$ is said to be T -unstable if t is not T -semistable. We will write $\mathbb{C}[T^\vee]_\bullet^H$ for the ring of H -invariant functions on $\mathbb{P}T$. The *GIT quotient* of $\mathbb{P}T$ with respect to H is the scheme

$$\mathbb{P}T//H := \text{Proj}(\mathbb{C}[T^\vee]_\bullet^H).$$

By construction, the inclusion of rings $\mathbb{C}[T^\vee]_\bullet^H \subset \mathbb{C}[T^\vee]_\bullet$ induces a rational map of projective spectra

$$\pi : \mathbb{P}T \dashrightarrow \mathbb{P}T//H$$

which is defined at a point $t \in \mathbb{P}T$ if and only if t is H -semistable. We introduce some notation:

Definition 3.1. Let $G = \text{SL}(U) \times \text{SL}(V) \times \text{SL}(W)$.

Definition 3.2. We will write

$$p_U : |\mathcal{O}_{\mathbb{P}U}(3)| \dashrightarrow |\mathcal{O}_{\mathbb{P}U}(3)|//\text{SL}(U)$$

for the quotient map.

For each labelling of $\{A, B\} = \{U, V, W\}$ we let π_A or π_{AB} or π_{UVW} be the resulting quotient map from $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ to its GIT quotient by $\text{SL}(A)$ or $\text{SL}(A) \times \text{SL}(B)$ or G . As examples:

$$\begin{aligned} \pi_U &: \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee) \dashrightarrow \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)//\text{SL}(U), \\ \pi_{VW} &: \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee) \dashrightarrow \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)//(\text{SL}(V) \times \text{SL}(W)). \end{aligned}$$

Fix identifications of $V \cong \mathbb{C}^3$ and $W \cong \mathbb{C}^3$. Taking the determinant $\det(\varphi_U)$ of a 3×3 matrix $\varphi \in U^\vee \otimes \text{Hom}(W, V^\vee) \cong U^\vee \otimes \text{Mat}_{3 \times 3}(\mathbb{C})$ gives a G -equivariant function

$$\det : U^\vee \otimes V^\vee \otimes W^\vee \longrightarrow \text{Sym}^3(U^\vee).$$

Other choices of bases for V and W only rescale the map $\det$ by a scalar, this choice will not matter after projectivisation. Pullback by $\det$ induces a G -equivariant map of graded rings

$$\det^* : \mathbb{C}[\text{Sym}^3(U)]_\bullet \longrightarrow \mathbb{C}[U \otimes V \otimes W]_\bullet.$$

Definition 3.3. By abuse of notation, we let

$$\det : \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee) \dashrightarrow |\mathcal{O}_{\mathbb{P}U}(3)| = \mathbb{P}\text{Sym}^3(U^\vee)$$

be the resulting G -equivariant map of projective spectra induced by $\det^*$. Similarly to Definition 3.2, for each labelling of $\{A, B\} \in \{U, V, W\}$ we let $\det_A$ and $\det_{AB}$ and $\det_{UVW}$ be the resulting map after quotienting both sides by $\text{SL}(A)$ or $\text{SL}(A) \times \text{SL}(B)$ or G . For example:

$$\begin{aligned} \det_V &: \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)//\text{SL}(V) \dashrightarrow |\mathcal{O}_{\mathbb{P}U}(3)|, \\ \det_{UVW} &: \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)//G \dashrightarrow |\mathcal{O}_{\mathbb{P}U}(3)|//\text{SL}(U). \end{aligned}$$

Note that for each choice of $\square = V, W, VW$, or omitting $\square$, we have $p_U \circ \det_\square = \det_{U\square} \circ \pi_{U\square}$.

Similarly, fixing a choice of bases $U \cong \mathbb{C}^4$ and $W \cong \mathbb{C}^3$ gives four G -equivariant functions

$$m_i : U^\vee \otimes V^\vee \otimes W^\vee \longrightarrow \text{Sym}^3(V^\vee),$$

where m_i outputs the i^{th} minor of the matrix φ_U . The four functions m_0, m_1, m_2, m_3 generate an ideal sheaf $\mathcal{I} \subset \mathcal{O}_{\mathbb{P}V \times \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)}$, independent of the choice of bases, which cuts out a family of subschemes $\mathcal{X} \subset \mathbb{P}V \times \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$. We write $\mathcal{X}^{\text{len } 6}$ for the restriction of $\mathcal{X}$ to the open subset $\mathbb{P}V \times \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)^{\text{len } 6}$ of points (v, φ) where the associated subscheme X_φ is a length 6 subscheme of $\mathbb{P}V$. The scheme $\mathcal{X}^{\text{len } 6}$ is flat over $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)^{\text{len } 6}$.

Definition 3.4. We let

$$\chi : \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee) \dashrightarrow \text{Hilb}_6(\mathbb{P}V)$$

be the morphism associated to the flat family $\mathcal{X}^{\text{len } 6} \rightarrow \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)^{\text{len } 6}$. We let

$$\chi_{UW} : \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee) // (\text{SL}(U) \times \text{SL}(W)) \dashrightarrow \text{Hilb}_6(\mathbb{P}V)$$

be the induced map from the GIT quotient.

By construction, we have $\chi(\varphi) = X_\varphi$ and $\chi = \chi_{UW} \circ \pi_{UW}$ when the right hand side is defined, noting that χ_{UW} is well-defined because χ is $\text{SL}(U) \times \text{SL}(W)$ -invariant.

The quotients of $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ by $\text{SL}(U)$ or $\text{SL}(V)$ or $\text{SL}(W)$ are the respective Grassmannians $\text{Gr}(4, V^\vee \otimes W^\vee) \cong \text{Gr}(4, 9)$ or $\text{Gr}(3, U^\vee \otimes W^\vee)$ or $\text{Gr}(3, U^\vee \otimes V^\vee)$, the latter two are isomorphic to $\text{Gr}(3, 12)$. All $\text{SL}(A)$ -semistable points are $\text{SL}(A)$ -stable and a tensor φ is $\text{SL}(A)$ -stable if and only if the corresponding map $\varphi_A : A \rightarrow B^\vee \otimes C^\vee$ is injective (where we relabel $\{A, B, C\} = \{U, V, W\}$).

The moduli space $\text{Gr}(3, U^\vee \otimes V^\vee)$ is a compactification of the moduli space of pairs (S, X) where $S \subset \mathbb{P}U$ is a canonical cubic surface and $X \subset \mathbb{P}V$ is an admissible subscheme associated to S . The moduli space $\text{Gr}(3, V^\vee \otimes W^\vee)$ is a compactification of the moduli space of pairs (X, Y) where $X \subset \mathbb{P}V$ and $Y \subset \mathbb{P}W$ are Gale-dual admissible length 6 subschemes.

The quotient $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee) // (\text{SL}(V) \times \text{SL}(W))$ is the GIT moduli space of determinantal representations of cubic surfaces $S \subset \mathbb{P}U$. This quotient has been studied in great detail by Lehn, Lehn, Storger and van Straten, we refer the reader to [LLSS17] for all details as well as many closely related moduli spaces. In particular, they prove

Proposition 3.5. *Let $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ be a tensor.*

- *φ is $\text{SL}(V) \times \text{SL}(W)$ -stable if and only if $\varphi_U(U) \subset V^\vee \otimes W^\vee$ is not a linear subspace of block type $(1, 2)$ or $(2, 1)$. This occurs if and only if either:*
 - *the surface $S_\varphi \subset \mathbb{P}U$ is irreducible; or*
 - *the tensor φ is skew-symmetric.*
- *φ is $\text{SL}(V) \times \text{SL}(W)$ -semistable if and only if $\varphi_U(U) \subset V^\vee \otimes W^\vee$ is not a linear subspace of block type $(1, 3)$ or $(2, 2)$ or $(3, 1)$. The tensor φ is strictly $\text{SL}(V) \times \text{SL}(W)$ -semistable if and only if the surface $S_\varphi \subset \mathbb{P}U$ is reducible but φ is not skew-symmetric.*
- *φ is $\text{SL}(V) \times \text{SL}(W)$ -unstable if and only if both $\det(\varphi_U) = 0$ and φ is not skew-symmetric.*

Proof. Everything follows from Corollary 3.7 of [LLSS17], which we have restated in slightly more detail using our notation. $\square$

By Subsection 10.5 of [ABCH13], the quotients of $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ by $\text{SL}(U) \times \text{SL}(V)$ and $\text{SL}(U) \times \text{SL}(W)$ are the final minimal models of the respective Hilbert schemes $\text{Hilb}_6(\mathbb{P}W)$ and $\text{Hilb}_6(\mathbb{P}V)$. The quotient $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee) // (\text{SL}(U) \times \text{SL}(W))$ is an orbit space. The birational contraction

$$\gamma : \text{Hilb}_6(\mathbb{P}V) \dashrightarrow \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee) // (\text{SL}(U) \times \text{SL}(W))$$

sends a length 6 subscheme X to the $\text{SL}(U) \times \text{SL}(W)$ -orbit of $\text{SL}(U) \times \text{SL}(W)$ -stable tensors $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ such that $X \subset X_\varphi$ (which is defined whenever this orbit exists and is unique).

In Subsection 3.1 we give some background on Gale-duality and use it to construct the map γ . We give a new proof that γ is a birational contraction in Subsection 3.2 and show that γ is an isomorphism when restricted to the locus of admissible length 6 subscheme; we use this isomorphism to prove our classification of $\text{SL}(U) \times \text{SL}(V) \times \text{SL}(W)$ -semistability and stability in Section 4.

3.1. Gale-duality. There is an involution, known as the *Gale transform*, that sends a sufficiently general ordered set of $r + s + 2$ points $X \subset \mathbb{P}^r$ to another ordered set of $r + s + 2$ points $Y \subset \mathbb{P}^s$, uniquely defined up to $\mathrm{PGL}(s)$. It is an involution in the sense that the Gale transform of Y is projectively equivalent to the original point configuration X . This was first noticed by Coble in 1922 [Cob22] and has a simple linear-algebraic description: two ordered sets of $r + s + 2$ points $\{x_1, \dots, x_{r+s+2}\} \subset \mathbb{P}^r$ and $\{y_1, \dots, y_{r+s+2}\} \subset \mathbb{P}^s$ are said to be *Gale-dual* if the set of tensor products $\{x_i \otimes y_j\}_{i=1}^{r+s+2} \subset \mathbb{P}(\mathbb{C}^{r+1} \otimes \mathbb{C}^{s+1})$ is linearly dependent.

Eisenbud and Popescu extended the Gale transform to Gorenstein subschemes in [EP99; EP00]. We give a sketch of their construction, all technical details and related results can be found in Sections 5 and 6 of [EP00]. In most cases, the Gale transform of a subscheme $X \subset \mathbb{P}^r$ is determined by the structure of the free resolution of the ideal sheaf $\mathcal{I}_X \subset \mathcal{O}_{\mathbb{P}^r}$. In the case that $r = s = 2$ and $X \subset \mathbb{P}^2$ is sufficiently general (e.g. admissible), the pair of the subscheme $X \subset \mathbb{P}^2$ and its Gale transform $Y \subset \mathbb{P}^2$ also reconstruct the free resolution of $\mathcal{I}_X$ given in Proposition 2.9.

Let $X \subset \mathbb{P}V$ be a Gorenstein length 6 subscheme not contained in a line, with dualising sheaf ω_X . The composition of the trace map $\tau : H^0(X, \omega_X) \rightarrow \mathbb{C}$ with the cup product induces the perfect pairing

$$H^0(X, \mathcal{O}_X(1)) \otimes H^0(X, \omega_X(-1)) \longrightarrow H^0(X, \omega_X) \xrightarrow{\tau} \mathbb{C}$$

of Serre duality. We let $V'^\vee = (V^\vee)^\perp \subset H^0(X, \omega_X(-1))$ be the annihilator of $V^\vee \subset H^0(X, \mathcal{O}_X(1))$ with respect to this pairing; we say that the pair of linear series $V^\vee \subset H^0(X, \mathcal{O}_X(1))$ and $V'^\vee \subset H^0(X, \omega_X(-1))$ are *Gale-dual*.

Definition 3.6. Let $X \subset \mathbb{P}V$ be a Gorenstein length 6 subscheme that is not contained in a line. The image $X' \subset \mathbb{P}V'$ of X under the linear series $V'^\vee \subset H^0(X, \omega_X(-1))$, as above, is called the *Gale transform* of the scheme $X \subset \mathbb{P}V$.

If a projective isomorphism $\mathbb{P}V' \rightarrow \mathbb{P}W$ induces an isomorphism from $X' \subset \mathbb{P}V'$ to a subscheme $Y \subset \mathbb{P}W$ then we say the subschemes $X \subset \mathbb{P}V$ and $Y \subset \mathbb{P}W$ are *Gale-dual*.

One can check that Definition 3.6 agrees with the previous linear-algebraic definition. The reason we require X to not be contained in a line is to ensure the map $V^\vee \rightarrow H^0(X, \mathcal{O}_X(1))$ is injective. The Gale transform of a length 6 subscheme contained in a line can still be defined, but it lives on a rational normal curve inside $\mathbb{P}^3$.

To a Gorenstein length 6 subscheme $X \subset \mathbb{P}V$ we associate the linear map

$$\phi : V^\vee \otimes V'^\vee \longrightarrow \ker(\tau) \subset H^0(X, \omega_X).$$

The vector space $\ker(\tau)$ is 5-dimensional and, under the additional assumption that ϕ is surjective, we consider the inclusion of its 4-dimensional kernel

$$\psi_X : \ker(\phi) \rightarrow V^\vee \otimes V'^\vee.$$

A sufficient condition for surjectivity of ϕ is that X contains a length 4 subscheme $X' \subset X$ in linearly general position (the intersection $L \cap X'$ of X' with any line $L \subset \mathbb{P}V$ has length at most 2) [EP00, Proposition 5.8].

Definition 3.7. Let $X \subset \mathbb{P}V$ be a Gorenstein length 6 subscheme. Following the above construction, when the map $\phi : V^\vee \otimes V'^\vee \rightarrow \ker(\tau)$ is surjective we call the tensor $\psi_X \in \mathrm{Hom}(\ker(\phi), V^\vee \otimes V'^\vee)$ the *associated tensor* to X .

We have $\dim \ker(\phi) = 4$ and $\dim V' = 3$ so, after choosing identifications $\ker(\phi) \cong U$ and $V' \cong W$, the tensor ψ can be identified with a $4 \times 3 \times 3$ tensor $\psi \in U^\vee \otimes V^\vee \otimes W^\vee$. In certain

cases, the associated tensor ψ to a Gorenstein length 6 subscheme $X \subset \mathbb{P}V$ recovers the original scheme X as X_ψ .

For a tensor $\varphi \in U^\vee \otimes V^\vee \otimes W^\vee$ we let $\tilde{\varphi}_V^\vee : U \otimes \mathcal{O}_{\mathbb{P}V}(1) \rightarrow W^\vee \otimes \mathcal{O}_{\mathbb{P}V}(2)$ be the map of sheaves presented by the matrix $\varphi_V \in V^\vee \otimes \text{Hom}(U, W^\vee)$. We have

Proposition 3.8. *Let $\varphi \in U^\vee \otimes V^\vee \otimes W^\vee$ be a tensor. If the associated subscheme $X_\varphi \subset \mathbb{P}V$ is finite and the map $\varphi_V : V \rightarrow U^\vee \otimes W^\vee$ is injective then X_φ is length 6 (the same applies for Y_φ and φ_W). The following are equivalent:*

- (i) X_φ and Y_φ are finite.
- (ii) X_φ is finite and Gorenstein.
- (iii) Y_φ is finite and Gorenstein.

If one of the above three statements holds then the schemes X_φ and Y_φ are Gale-dual.

Proof. Everything follows from Theorem 6.1 and Proposition 6.2 of [EP00], except the claim that a finite subscheme is length 6. If φ_V is injective then φ_V induces a linear embedding of $\mathbb{P}V \xrightarrow{\sim} \mathbb{P}(\varphi_V(V)) \subset \mathbb{P}(U^\vee \otimes W^\vee)$. The subscheme X_φ is the scheme-theoretic intersection of $\mathbb{P}V$ with the rank at most 2 subvariety of $\mathbb{P}(U^\vee \otimes W^\vee)$, which is degree 6 and codimension 2. This intersection is then length 6 whenever it is finite. $\square$

The case for Gorenstein subschemes contained in a conic is slightly different. While we cannot use the Hilbert-Burch theorem to construct a determinantal representation φ for a length 6 Gorenstein subscheme $X \subset \mathbb{P}V$ contained in a conic, the construction of Definition 3.7 still produces an associated tensor well-defined whenever X is the complete intersection of a conic and a cubic. In this case, the associated subscheme $X_\varphi \subset \mathbb{P}V$ to φ is the conic containing X . The following construction is essentially a reformulation of the proof of Theorem 7.1 of [EP00]. In particular, Eisenbud and Popescu prove that when $X \subset \mathbb{P}V$ is a Gorenstein length 6 subscheme contained in a conic, its Gale transform $X' \subset \mathbb{P}^2$ is isomorphic to X as a subscheme of $\mathbb{P}^2$.

Remark 3.9. Concretely, let $X \subset \mathbb{P}V$ be the complete intersection of a conic Q and a cubic curve. The scheme X satisfies condition (b) of Theorem 7.1 of [EP00], in particular there exists an isomorphism $\alpha' : \mathcal{O}_X(1) \rightarrow \mathcal{O}_X(1)^\vee = \omega_X(-1)$ such that the composition

$$V^\vee \longrightarrow H^0(X, \mathcal{O}_X(1)) \xrightarrow{\alpha'} H^0(X, \mathcal{O}_X(1)^\vee) \longrightarrow V$$

is zero. The isomorphism α' is a twist of the isomorphism $\alpha : \mathcal{O}_X(2) \rightarrow \omega_X$ given by the adjunction formula. Now, consider the exact sequence of cohomology

$$0 \longrightarrow H^0(\mathbb{P}V, \mathcal{I}_X(2)) \xrightarrow{f} H^0(\mathbb{P}V, \mathcal{O}_{\mathbb{P}V}(2)) \xrightarrow{g} H^0(X, \mathcal{O}_X(2)) \xrightarrow{h} H^1(\mathbb{P}V, \mathcal{I}_X(2)) \longrightarrow 0$$

where g sends $Q \mapsto 0$ (viewing $Q \in \text{Sym}^2(V^\vee)$ as a quadratic form by abuse of notation). We identify $\text{coker}(f) = \text{Sym}^2(V^\vee)/\text{Span}\{Q\}$ with $\ker h \subset H^0(X, \mathcal{O}_X(2))$. The map h generates $\text{Hom}_{\mathbb{C}}(H^0(X, \mathcal{O}_X(2)), \mathbb{C})$ as an $\mathcal{O}_X$ -algebra. There is an isomorphism $H^1(\mathbb{P}V, \mathcal{I}_X(2)) \cong \mathbb{C}$ so that the following diagram commutes

$$\begin{array}{ccccc} H^0(X, \mathcal{O}_X(1)) \times H^0(X, \mathcal{O}_X(1)) & \xrightarrow{\sim} & H^0(X, \mathcal{O}_X(2)) & \xrightarrow{h} & H^1(\mathbb{P}V, \mathcal{I}_X(2)) \\ \text{id} \times \alpha' \downarrow & & \alpha \downarrow & & \downarrow \sim \\ H^0(X, \mathcal{O}_X(1)) \times H^0(X, \omega_X(-1)) & \xrightarrow{\sim} & H^0(X, \omega_X) & \xrightarrow{\tau} & \mathbb{C} \end{array}$$

and, as a result, we identify h with the trace map τ . In particular, $g(H^0(\mathbb{P}V, \mathcal{O}_{\mathbb{P}V}(2))) = \ker h$, i.e. h annihilates the image of $V^\vee \otimes V^\vee$ inside $H^0(X, \mathcal{O}_X(2))$. The linear map $V^\vee \otimes V^\vee \rightarrow \ker h$ can then be identified with the natural map

$$\phi : V^\vee \otimes V^\vee \longrightarrow \text{Sym}^2(V^\vee) / \text{Span}\{Q\}$$

and $\ker \phi = \text{Span}\{Q\} \oplus \wedge^2 V^\vee$. We then consider the inclusion

$$(5) \quad \psi : \text{Span}\{Q\} \oplus \wedge^2 V^\vee \longrightarrow V^\vee \otimes V^\vee.$$

We choose isomorphisms $U \cong \text{Span}\{Q\} \oplus \wedge^2 V^\vee$ and $W^\vee \cong V^\vee$ (only onto the second copy of $V^\vee$), which identifies ψ with a rank $\text{rk}(Q)$ skew-symmetric tensor $\varphi \in \Sigma_{\text{rk}(Q)}$, unique up to $\text{GL}(U) \times \text{GL}(W)$ -equivalence.

The tensor φ is the tensor associated to the subscheme X , in the sense of Definition 3.7. In coordinates, φ is $\text{GL}(U)$ -equivalent to the skew-symmetric tensor

$$\varphi_U = u_0 Q + \begin{pmatrix} 0 & u_3 & -u_2 \\ -u_3 & 0 & u_1 \\ u_2 & -u_1 & 0 \end{pmatrix}.$$

The adjoint 4×3 tensor $\varphi_V \in V^\vee \otimes \text{Hom}(W, U^\vee)$ to φ_U is

$$\varphi_V = \begin{pmatrix} q_0 & q_1 & q_2 \\ 0 & -v_2 & v_1 \\ v_2 & 0 & -v_0 \\ -v_1 & v_0 & 0 \end{pmatrix},$$

for some elements $q_0, q_1, q_2 \in V^\vee$. There is a unique isomorphism $W^\vee \cong V^\vee$ that identifies the matrix $(q_0 \ q_1 \ q_2) \in V^\vee \otimes W^\vee$ with the symmetric matrix $Q \in \text{Sym}^2(V^\vee) \subset V^\vee \otimes V^\vee$. The 3×3 minors of φ_V are

$$(0, v_0(q_0 v_0 + q_1 v_1 + q_2 v_2), v_1(q_0 v_0 + q_1 v_1 + q_2 v_2), v_2(q_0 v_0 + q_1 v_1 + q_2 v_2)).$$

These minors generate the ideal sheaf $\mathcal{I}_Q$ in degree 3, so $X_\varphi = Q$ is the conic containing the original scheme X .

When $h^0(\mathbb{P}V, \mathcal{I}_X(2)) \geq 2$ this procedure fails. Concretely, we only have $g(H^0(\mathbb{P}V, \mathcal{O}_{\mathbb{P}V}(2))) \subset \ker h$. The above procedure produces a tensor

$$\psi : H^0(\mathbb{P}V, \mathcal{I}_X(2)) \oplus \wedge^2 V^\vee \longrightarrow V^\vee \otimes V^\vee$$

and there is a family of associated tensors to the length 6 subscheme X parametrised by the possible choices of injections $U \rightarrow H^0(\mathbb{P}V, \mathcal{I}_X(2)) \oplus \wedge^2 V^\vee$.

Remark 3.10. The map sending a Gorenstein subscheme $X \subset \mathbb{P}V$ to its associated tensor can also be done in families. For ease of notation we only write the construction over affine schemes. Let $\text{Spec } A$ be an affine scheme and fix a subscheme $X_A \subset \mathbb{P}V \times \text{Spec } A$, flat and Gorenstein over A , whose fibres $X_{\mathfrak{a}} \subset \mathbb{P}V \times \text{Spec}(A/\mathfrak{a})$ are all length 6 and span the projective space $\mathbb{P}V \times \text{Spec}(A/\mathfrak{a})$. In particular, X_A is A -flat. We let $\tau : H^0(X_A, \omega_{X_A}) \rightarrow A$ be the trace map.

Because each $X_{\mathfrak{a}}$ spans $\mathbb{P}V \times \text{Spec}(A/\mathfrak{a})$, the map $f : V^\vee \otimes A \rightarrow H^0(X_A, \mathcal{O}_{X_A}(1))$ is injective. We then identify $V^\vee \otimes A$ with its image $f(V^\vee \otimes A) \subset H^0(X_A, \mathcal{O}_{X_A}(1))$. The annihilator of $V^\vee \otimes A$ with respect to the trace pairing

$$H^0(X_A, \mathcal{O}_{X_A}(1)) \otimes_A H^0(X_A, \omega_{X_A}(-1)) \longrightarrow A$$

is then a locally free submodule $\mathcal{W}^\vee \subset H^0(X_A, \omega_{X_A}(-1))$ of rank 3. We let $Y_A \subset \mathbb{P}\mathcal{W}$ be the image of X_A under the linear series $(\mathcal{W}^\vee, \omega_{X_A}(-1))$. For each maximal ideal $\mathfrak{a} \in \operatorname{Spec} A$ the pair of subschemes $X_{\mathfrak{a}} \subset \mathbb{P}V \times \operatorname{Spec}(A/\mathfrak{a})$ and $Y_{\mathfrak{a}} \subset \mathbb{P}\mathcal{W} \times \operatorname{Spec}(A/\mathfrak{a})$ are Gale-dual.

As in the construction preceding Definition 3.7, Serre duality gives a linear map

$$\phi : V^\vee \otimes \mathcal{W}^\vee \longrightarrow \ker(\tau) \subset H^0(X_A, \omega_{X_A}).$$

We let $\mathcal{U} := \ker(\phi)$ and consider its inclusion

$$\psi : \mathcal{U} \longrightarrow V^\vee \otimes \mathcal{W}^\vee.$$

Under the additional assumption that each fibre $X_{\mathfrak{a}}$ contains a length 4 subscheme $X'_{\mathfrak{a}} \subset X_{\mathfrak{a}}$ in linearly general position, the map $\phi \otimes \operatorname{id}_{A/\mathfrak{a}} : (V^\vee \otimes \mathcal{W}^\vee) \otimes_A A/\mathfrak{a} \rightarrow \ker(\tau) \otimes_A A/\mathfrak{a}$ of locally free A -modules is surjective for each maximal ideal $\mathfrak{a} \in \operatorname{Spec} A$; so ϕ is surjective. Then $\mathcal{U}$ is a rank 4 locally free A -module. The map ψ can be identified with a section

$$\psi' : \operatorname{Spec} A \longrightarrow \mathbb{P}(\mathcal{U}^\vee \otimes V^\vee \otimes \mathcal{W}^\vee)$$

into a projectivised vector bundle of $4 \times 3 \times 3$ tensors.

3.2. The final minimal model of the Hilbert scheme of 6 points in $\mathbb{P}^2$. The construction of Remark 3.10 induces a rational map

$$\gamma : \operatorname{Hilb}_6(\mathbb{P}V) \dashrightarrow \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee) // (\operatorname{SL}(U) \times \operatorname{SL}(W)).$$

By [ABCH13], GIT semistability of $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ under the $\operatorname{SL}(U) \times \operatorname{SL}(W)$ -action coincides with a Bridgeland stability condition on $D^b \operatorname{Coh}(\mathbb{P}V)$. The map γ is a birational contraction contracting the divisor $C_6 \subset \operatorname{Hilb}_6(\mathbb{P}V)$ that parametrises length 6 subschemes $X \subset \mathbb{P}V$ contained in a conic. Details can be found in Section 8 and Subsection 10.5 of their paper, however they do not construct this contraction. The purpose of Subsection 3.2 is to explicitly construct the contraction γ , we will use the map γ to compute orbit closures in Section 4.

The map γ is defined when the following three conditions hold:

- (i) the scheme $X \subset \mathbb{P}V$ is Gorenstein; and
- (ii) the map $V^\vee \otimes W^\vee \rightarrow \ker(\tau) \subset H^0(X, \omega_X)$ is surjective; and
- (iii) the associated tensor φ is $\operatorname{SL}(U) \times \operatorname{SL}(W)$ -semistable.

The $\operatorname{SL}(U) \times \operatorname{SL}(W)$ -semistability of $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ is known, a result of Dr  zet [Dr  87]:

Proposition 3.11. *A tensor $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ is $\operatorname{SL}(U) \times \operatorname{SL}(W)$ -semistable if and only if φ_V is not $\operatorname{SL}(U) \times \operatorname{SL}(W)$ -equivalent to a matrix of one of the following forms:*

$$\begin{pmatrix} * & * & * \\ * & * & * \\ * & * & * \\ 0 & 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} * & * & * \\ * & * & * \\ * & 0 & 0 \\ * & 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} * & * & * \\ * & * & 0 \\ * & * & 0 \\ * & * & 0 \end{pmatrix}.$$

All $\operatorname{SL}(U) \times \operatorname{SL}(W)$ -semistable points are $\operatorname{SL}(U) \times \operatorname{SL}(W)$ -stable.

A quick computation of the 3×3 minors shows that when φ is $\operatorname{SL}(U) \times \operatorname{SL}(W)$ -unstable ($\operatorname{SL}(U) \times \operatorname{SL}(W)$ -equivalent to one of the above three matrices) then the ideal cutting out the subscheme $X_\varphi \subset \mathbb{P}V$ is generated by either:

- (i) a cubic polynomials; or
- (ii) two terms $l_1 q$ and $l_2 q$, where l_1 and l_2 are linear and q is a quadratic; or
- (iii) three terms $l q_1, l q_2, l q_3$, where l is linear and q_1, q_2, q_3 are quadratics.

Some (or all) of these generators may be zero. In particular, if $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ is $\mathrm{SL}(U) \times \mathrm{SL}(W)$ -unstable then the scheme X_φ contains a positive-dimensional component, however the converse is not true. Some sufficient conditions for $\mathrm{SL}(U) \times \mathrm{SL}(W)$ -semistability include

Proposition 3.12. *A tensor $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ is $\mathrm{SL}(U) \times \mathrm{SL}(W)$ -semistable if:*

- (i) *the scheme X_φ is finite; or*
- (ii) *the tensor φ is a positive rank skew-symmetric tensor ($\varphi \in \Sigma \setminus \Sigma_0$).*

Proof. Only (ii) is unproven, this is immediate because a positive rank skew-symmetric tensor $\varphi \in \Sigma \setminus \Sigma_0$ does not satisfy the block conditions of Proposition 3.11. $\square$

Remark 3.13. The GIT moduli space $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee) // (\mathrm{SL}(U) \times \mathrm{SL}(W))$ is a moduli space of sheaves. Concretely, $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ parametrises complexes of the form

$$(6) \quad W \otimes \mathcal{O}_{\mathbb{P}V}(-4) \xrightarrow{\tilde{\varphi}_V} U^\vee \otimes \mathcal{O}_{\mathbb{P}V}(-3)$$

inside $D^b \mathrm{Coh}(\mathbb{P}V)$ (forgetting the basis for U and W). A complex of the form of (6) is quasi-isomorphic to the complex

$$(7) \quad \ker(\tilde{\varphi}_V) \longrightarrow 0 \longrightarrow 0 \longrightarrow \mathrm{coker}(\tilde{\varphi}_V),$$

which is the complex associated to the sheaf $\mathrm{coker}(\tilde{\varphi}_V)$ whenever $\ker(\tilde{\varphi}_V) = 0$. The kernel $\ker(\tilde{\varphi}_V)$ is non-zero if and only if $\varphi_V(V) \subset U^\vee \otimes W^\vee$ is a rank at most 2 subspace. This is only possible if either $\varphi_V(V)$ is of block type (2, 3) or (3, 2) or (4, 1), and hence $\mathrm{SL}(U) \times \mathrm{SL}(W)$ -unstable, or if $\varphi_V(V)$ is a skew-symmetric subspace. But then $\varphi_V(V)$ is of block type (1, 3) and is also $\mathrm{SL}(U) \times \mathrm{SL}(W)$ -unstable. So all points in $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)^{\mathrm{SL}(U) \times \mathrm{SL}(W)-ss}$ correspond to complexes that are quasi-isomorphic to sheaves.

Because all $\mathrm{SL}(U) \times \mathrm{SL}(W)$ -semistable points are stable, $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee) // (\mathrm{SL}(U) \times \mathrm{SL}(W))$ is an orbit space and its points parametrise complexes quasi-isomorphic to a complex of the form (7) with $\ker(\tilde{\varphi}_V) = 0$.

Recall that Σ is the $\mathrm{SL}(U) \times \mathrm{SL}(W)$ -orbit of tensors of the form

$$\varphi_U = u_0 Q + \begin{pmatrix} 0 & u_3 & -u_2 \\ -u_3 & 0 & u_1 \\ u_2 & -u_1 & 0 \end{pmatrix},$$

where the skew-symmetric part of a tensor $\varphi \in \Sigma$ gives an identification $V \cong W$. The matrix $Q \in V^\vee \otimes W^\vee$ can be chosen to be symmetric with respect to this identification. The subscheme X_φ is cut out by the quadratic form $v^T Q v = 0$, and as a result we have

Proposition 3.14. *The map $\varphi \mapsto X_\varphi$ induces an isomorphism $\Sigma // (\mathrm{SL}(U) \times \mathrm{SL}(W)) \cong |\mathcal{O}_{\mathbb{P}V}(2)|$.*

Proof. The family $\mathcal{Q} \subset \mathbb{P}V \times (\Sigma \setminus \Sigma_0)$ of associated conics $X_\varphi \subset \mathbb{P}V$ to tensors $\varphi \in \Sigma \setminus \Sigma_0$ is flat over $\Sigma \setminus \Sigma_0$. So $\mathcal{Q}$ corresponds to an $\mathrm{SL}(U) \times \mathrm{SL}(W)$ -invariant morphism $f : \Sigma \setminus \Sigma_0 \rightarrow |\mathcal{O}_{\mathbb{P}V}(2)|$. The morphism f is surjective. We have $\Sigma^{\mathrm{SL}(U) \times \mathrm{SL}(W)-ss} = \Sigma \setminus \Sigma_0$, so f factors through the GIT quotient and induces a surjective morphism $f_{UV} : \Sigma // (\mathrm{SL}(U) \times \mathrm{SL}(W)) \rightarrow |\mathcal{O}_{\mathbb{P}U}(2)|$. Two tensors $\varphi, \varphi' \in \Sigma$ are $\mathrm{SL}(U) \times \mathrm{SL}(W)$ -equivalent if and only if their associated quadratic forms A and A' are equal, so f is a bijective morphism.

Finally, the $\mathrm{PGL}(U) \times \mathrm{PGL}(W)$ -stabiliser of every tensor $\varphi \in \Sigma$ is trivial, so the GIT quotient $\Sigma // (\mathrm{PGL}(U) \times \mathrm{PGL}(W))$ is smooth. The two GIT quotients $\Sigma // (\mathrm{SL}(U) \times \mathrm{SL}(W))$ and $\Sigma // (\mathrm{PGL}(U) \times \mathrm{PGL}(W))$ are isomorphic as schemes (the centre of $\mathrm{SL}(U) \times \mathrm{SL}(W)$ acts trivially),

so $\Sigma//(\mathrm{SL}(U) \times \mathrm{SL}(W))$ is smooth. Then the morphism f is a bijective morphism of smooth varieties, so is an isomorphism. $\square$

Definition 3.15. Let $C_6 \subset \mathrm{Hilb}_6(\mathbb{P}V)$ be the divisor parametrising length 6 subschemes $X \subset \mathbb{P}V$ contained in a conic.

Let $\mathcal{H} \subset \mathrm{Hilb}_6(\mathbb{P}V)$ be the open subvariety parametrising determinantal, Gorenstein length 6 subschemes $X \subset \mathbb{P}V$ that are not contained in a conic.

Proposition 3.16. *The map χ_{UW} is the birational inverse of the map*

$$\gamma : \mathrm{Hilb}_6(\mathbb{P}V) \dashrightarrow \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee) // (\mathrm{SL}(U) \times \mathrm{SL}(W)).$$

The map γ contracts C_6 onto $\Sigma//(\mathrm{SL}(U) \times \mathrm{SL}(W)) \cong |\mathcal{O}_{\mathbb{P}V}(2)|$. The restriction of γ to $\mathcal{H}$ is an isomorphism onto its image.

Proof. The fact that γ is a birational contraction contracting C_6 is already known, a result of [ABCH13]. Explicitly, the contraction comes from the construction of Remarks 3.9 and 3.10.

To show γ is an isomorphism when restricted to $\mathcal{H}$, we need to show that $\chi_{UW} \circ \gamma = \mathrm{id}$ whenever γ is defined. Because both $\mathrm{Hilb}_6(\mathbb{P}V)$ and $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee) // (\mathrm{SL}(U) \times \mathrm{SL}(W))$ are smooth then it suffices to show that γ is bijective when restricted to $\mathcal{H}$. Let $[X] \in \mathcal{H}$ be a determinantal, Gorenstein length 6 subscheme. Then $X = X_\varphi$ for some $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)^{\mathrm{SL}(U) \times \mathrm{SL}(W)-ss}$. By [EP00, Theorem 6.1] we have $(\chi_{UW} \circ \gamma)([X_\varphi]) = \chi(\varphi) = [X_\varphi]$ as desired. $\square$

4. THE FULL $4 \times 3 \times 3$ GIT QUOTIENT

Recall that $G = \mathrm{SL}(U) \times \mathrm{SL}(V) \times \mathrm{SL}(W)$. The group G is a finite extension of the symmetry group $\mathrm{PGL}(U) \times \mathrm{PGL}(V) \times \mathrm{PGL}(W)$ of $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$. The primary goal of this paper is to describe the G -semistability and G -stability of the moduli space $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ of $4 \times 3 \times 3$ tensors. To describe the G -semistable and G -stable loci of $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ we will need to classify all of the points $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$. In particular, we need a detailed enumeration of all canonical tensors; this was done by Ng in [Ng02], we review their classification and describe the resulting stratification of $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ in Subsection 4.1.

Subsections 4.2, 4.3 and 4.4 are devoted to describing the G -semistable, G -stable, and G -polystable points of $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$. We will give sufficient geometric conditions for G -semistability and G -stability and then prove these conditions are necessary using the Hilbert-Mumford criterion. Along the way we describe the G -invariant ring $\mathbb{C}[U \otimes V \otimes W]_\bullet^G$ up to finite extension and give a detailed description of a stratification of the G -semistable locus $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)^{G-ss}$, two necessary ingredients for our proofs of G -stability and G -polystability.

4.1. A classification of $4 \times 3 \times 3$ tensors. It is convenient to stratify the set of tensors $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ by subsets corresponding to the singularity type that the associated cubic surface S_φ has. We have

Proposition 4.1. *Every tensor $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ is of exactly one of the following types:*

- (i) φ is canonical (the associated surface S_φ is canonical); or
- (ii) S_φ is reducible; or
- (iii) S_φ is a cone over a cubic curve; or
- (iv) S_φ is non-normal, non-conical, and irreducible. In this case, the polynomial defining S is $\mathrm{SL}(U)$ -equivalent to either $u_2u_0^2 + u_3u_1^2$ or $u_2u_0^2 + u_3u_0u_1 + u_1^3$. Or
- (v) $\det(\varphi) = 0$ and $S_\varphi = \mathbb{P}U$.

Proof. We apply the classification of Proposition 2.1 to $\det(\varphi)$. The polynomial forms of (iv) can be found in [BW79]. $\square$

Ng listed almost all G -equivalence classes of tensors $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ in [Ng02] with the goal of eventually describing the GIT quotient $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)/G$, however the latter paper was never published. We draw heavily on their complete enumeration of canonical tensors for the remainder of the paper. We derive our own results for non-canonical tensors as their classification is unfortunately incomplete – it omits rank 2 and 3 skew-symmetric tensors, which are G -equivalent to

$$\varphi_U = u_0 Q + \begin{pmatrix} 0 & v_2 & -v_1 \\ -v_2 & 0 & v_0 \\ v_1 & -v_0 & 0 \end{pmatrix}$$

where Q is a rank 2 or 3 matrix. To enumerate all canonical tensors, [Ng02] uses the correspondence between determinantal representations of canonical cubic surfaces $S \subset \mathbb{P}U$ and admissible length 6 subschemes $X \subset \mathbb{P}V$ (recall Proposition 2.13). Explicitly, they use Maple to enumerate all $\mathrm{SL}(V)$ -equivalence classes of admissible length 6 subschemes $X \subset \mathbb{P}V$ and then compute the 4×3 presentation matrix for the minimal free resolution of the ideal sheaf $\mathcal{I}_X$.

Recall the subvarieties $\mathcal{S}(F) \subset |\mathcal{O}_{\mathbb{P}U}(3)|$, where $\mathcal{S}(F)$ is the set of cubic surfaces with F -type singularities. The image of the rational map

$$\det : \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee) \dashrightarrow |\mathcal{O}_{\mathbb{P}U}(3)|$$

is the subvariety $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee) \setminus \mathcal{S}(E_6)$ (recall that the E_6 cubic surface has no determinantal representations).

Definition 4.2. We let $\mathbb{S} := \det^{-1}(\mathcal{S}(\emptyset))$ be the set of determinantal representations of smooth cubic surfaces, it is open inside $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$. For every other strict Dynkin subdiagram $\emptyset \neq F \subsetneq \tilde{E}_6$ the preimage $\det^{-1}(\mathcal{S}(F))$ of $\mathcal{S}(F)$ splits into the disjoint union of a small number of (at most 6) connected components. We label these connected components $F(n)$ with a Latin letter $n \in \{a, b, c, d, e, f\}$.

This labelling is consistent with Ng's notation, we refer the reader to [Ng02] for a complete description of the points of each subvariety $F(n)$.

We use slightly different notation for $F = D_4$ to be consistent with [Ng02]. The preimage $\det^{-1}(\mathcal{S}(D_4))$ is connected, but Ng decomposes the preimage as the disjoint union of two components $D_4(a) \cup D_4(b) := \det^{-1}(\mathcal{S}(D_4))$. This decomposition is consistent with the notation of Proposition 2.4, in the sense that the associated cubic surface S_φ to a tensor $\varphi \in D_4(n)$ is isomorphic to $S_{D_4(n)}$. The subvarieties $D_4(a)$ and $D_4(b)$ are both G -orbits and $D_4(b) \subset \overline{D_4(a)}$. Each of the two isomorphism classes of D_4 cubic surfaces have a unique determinantal representation (up to G -equivalence). The technicalities of the determinantal representations of the D_4 cubic surfaces are unimportant for our results, we will see that all determinantal representations of a D_4 cubic surface are G -unstable.

Example 4.3. The subvariety $A_1 A_2(a) \subset \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ is the G -orbit of tensors of the form

$$\varphi_U = \begin{pmatrix} \lambda(\lambda+1)u_1 + (\lambda+1)u_3 & u_3 & u_0 \\ \lambda u_1 + u_3 & -u_1 & u_3 \\ u_2 & 0 & u_3 \end{pmatrix},$$

parametrised by $\lambda \in \mathbb{C} \setminus \{0, 1\}$.

Recall that $\chi : \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee) \dashrightarrow \text{Hilb}_6(\mathbb{P}V)$ is the rational map that sends a tensor φ to its associated length 6 subscheme $X_\varphi \subset \mathbb{P}V$, defined when X_φ is length 6. Using Propositions 2.13 and 3.16, the subvarieties $F(n)$ can be equivalently described in terms of their associated length 6 subschemes. The image $\chi(F(n)) \subset \text{Hilb}_6(\mathbb{P}V)$ is a locally closed subvariety parametrising canonical length 6 subschemes satisfying certain geometric conditions (in terms of the number and type of non-reduced points and the ways in which they intersect with lines).

Example 4.4. The subvariety $\chi(A_1 A_2(a)) \subset \text{Hilb}_6(\mathbb{P}V)$ parametrises length 6 subschemes $X \subset \mathbb{P}V$ satisfying (a) the following closed conditions:

- X contains a length 2 point $X_1 \subset X$; and
- X contains two length 3 subschemes $X_2, X_3 \subset X$ such that X_2 and X_3 are each contained in a line; and
- the subschemes X_2 and X_3 satisfy $X_1 \subset X_2$ and $(X_1)_{\text{red}} \subset X_3$.

and (b) the following open conditions:

- X is supported on at least five points; and
- X contains at most two length 3 collinear subschemes; and
- X contains no length 4 collinear subscheme; and
- X is not contained in a conic.

The conditions defining each $\chi(F(n))$ are similar to those of Example 4.4 and are straightforward to work out from the pictorial descriptions in [Ng02]. Formally, we decompose the open subset $\text{Hilb}_6(\mathbb{P}V)^{\text{adm}} \subset \text{Hilb}_6(\mathbb{P}V)$ parametrising admissible length 6 subschemes into locally closed subvarieties $\mathcal{H}_F$ indexed by the associated Dynkin diagram F to a subscheme $X \subset \mathbb{P}V$ (where F is the intersection graph of the (-2) -curves on the minimal resolution of singularities of the blowup $\text{Bl}_X \mathbb{P}V$). For each Dynkin diagram $F \subsetneq \tilde{E}_6$ the subvariety $\mathcal{H}_F$ decomposes as the disjoint union of a small number of connected components (which are $\chi(F(n))$), and one can check that each of these connected components are cut out by conditions similar to those of Example 4.4 (open and closed conditions in terms of the number and type of non-reduced points and how the point configuration intersects with lines, as well as the open condition that X is not contained in a conic). The conditions defining $D_4(a)$ and $D_4(b)$ are slightly different, but this technicality is unimportant for our results. When $F \neq D_4$ we have

Proposition 4.5. *For each stratum $F(n) \subset \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ where $F \neq D_4$ the induced map $\det|_{F(n)} : F(n) \rightarrow \mathcal{S}(F)$*

is surjective.

The fact that Proposition 4.5 doesn't apply to D_4 cubic surfaces is an artefact of our notation; every D_4 cubic surface has a determinantal representation and the set $D_4(a) \cup D_4(b)$ of determinantal representations of D_4 cubic surfaces is both connected and maps surjectively onto $\mathcal{S}(D_4)$.

In the moduli space of cubic surfaces we had $\mathcal{S}(F') \subset \overline{\mathcal{S}(F)}$ if and only if $F' \subset F$ is a Dynkin subdiagram. For tensors we have

Proposition 4.6. *Let $F, F' \subsetneq \tilde{E}_6$ be Dynkin subdiagrams and n, n' be Latin letters. A necessary condition for $F'(n') \subset \overline{F(n)}$ is that $F \subset F'$ as Dynkin diagrams.*

Proof. Suppose $F'(n') \subset \overline{F(n)}$. Then, because $\det : \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee) \dashrightarrow |\mathcal{O}_{\mathbb{P}U}(3)|$ is continuous and surjectively maps $F(n) \rightarrow \mathcal{S}(F)$ and $F'(n') \rightarrow \mathcal{S}(F')$, we have $\mathcal{S}(F') \subset \overline{\mathcal{S}(F)}$. This implies that $F' \subset F$. $\square$

The closure relations between the canonical strata of $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ can be most easily worked out by passing to the Hilbert scheme $\text{Hilb}_6(\mathbb{P}V)$. Concretely, we have

Proposition 4.7. *Let $F(n), F'(n') \subset \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ be strata. We have $F'(n') \subset \overline{F'(n')}$ if and only if $\chi(F'(n')) \subset \chi(F(n))$.*

Proof. Recall the commutative diagram

$$\begin{array}{ccc} \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee) & & \\ \chi \downarrow & \searrow \pi_{UW} & \\ \text{Hilb}_6(\mathbb{P}V) & \xrightarrow{\gamma} & \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee) / (\text{SL}(U) \times \text{SL}(W)). \end{array}$$

When restricted to the locus of canonical tensors/admissible subschemes, the maps χ and π_{UW} are morphisms and the map γ is an isomorphism onto its image. Because each stratum of type $F(n)$ is $\text{SL}(U) \times \text{SL}(W)$ -invariant then $F'(n') \subset \overline{F'(n')}$ if and only if $\pi_{UW}(F'(n')) \subset \overline{\pi_{UW}(F'(n'))}$. Passing through the isomorphism γ^{-1} and using the identity $\chi = \gamma^{-1} \circ \pi_{UW}$ (when the right hand side is defined) gives the desired result. $\square$

A corollary is that for any stratum $F'(n')$ and Dynkin diagram $F \subset F'$ there is a letter m such that $F'(n') \subset F(m)$ (this is straightforward to check by hand, although tedious, we don't use this fact).

By Lemma 3.4 of [LLSS17], the $\text{SL}(V) \times \text{SL}(W)$ -stabiliser of any canonical tensor is finite (because every canonical tensor is $\text{SL}(V) \times \text{SL}(W)$ -stable). Similarly, the $\text{SL}(U) \times \text{SL}(W)$ - and $\text{SL}(U) \times \text{SL}(V)$ -stabilisers are finite by Proposition 3.11. As a result, whenever $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ is a canonical tensor we have

$$(8) \quad \dim \text{Stab}_G(\varphi) = \dim \text{Stab}_{\text{SL}(U)}(S_\varphi) = \dim \text{Stab}_{\text{SL}(V)}(X_\varphi) = \dim \text{Stab}_{\text{SL}(W)}(Y_\varphi).$$

Applying orbit-stabiliser to (8) gives

$$\text{codim}(F(n) \subset \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)) = \text{codim}(S(F) \subset |\mathcal{O}_{\mathbb{P}U}(3)|).$$

The dimension of each stratum $F(n)$ follows from the dimensions of the stratum $S(F)$ listed in Proposition 2.4. We have

Proposition 4.8. *The subvariety $\mathbb{S} \subset \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$, parametrising tensors φ whose associated cubic surface S_φ is smooth, is dense inside $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ and $\dim \text{Stab}_G(\varphi) = 0$ whenever $\varphi \in \mathbb{S}$. More generally:*

- Let $F \subsetneq \tilde{E}_6$ be a Dynkin subdiagram where $F \neq 2A_2$ and $F \neq D_4$ and let n be a Latin letter. Then $\text{codim}(F(n)) = |F|$ and $\dim \text{Stab}_G(\varphi) = \min\{0, |F| - 4\}$ whenever $\varphi \in F(n)$. If additionally $|F| \geq 4$ then $F(n)$ is a G -orbit.
- Let $F = 2A_2$ and $n \in \{a, b\}$. Then $\text{codim}(2A_2(n)) = 3$ and $\dim \text{Stab}_G(\varphi) = 1$ whenever $\varphi \in 2A_2(n)$.
- Let $F = D_4$. Then $\text{codim}(D_4(a)) = 4$ and $\text{codim}(D_4(b)) = 5$. We have $\dim \text{Stab}_G(\varphi) = 0$ if $\varphi \in D_4(a)$ and $\dim \text{Stab}_G(\varphi) = 1$ if $\varphi \in D_4(b)$.

The subvariety $\Sigma_3 \subset \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ has codimension 7 and $\dim \text{Stab}_G(\varphi) = 3$ when $\varphi \in \Sigma_3$ is a rank 3 skew-symmetric tensor.

Proof. All that remains is to prove is that $\dim \text{Stab}_G(\varphi) = 3$ when $\varphi \in \Sigma_3$. There is a unique 3-dimensional subspace $U' \subset U$ such that $\varphi_U(U') \subset V^\vee \otimes W^\vee$. Recall from Definition 2.19 that there is a unique set of isomorphisms $U' \cong V \cong W \cong \mathbb{C}^3$ under which the map $\varphi_U|_{U'} : U' \rightarrow$

$V^\vee \otimes W^\vee$ is equivalent to the Hodge star map $\star : \mathbb{C}^3 \rightarrow \wedge^2(\mathbb{C}^3)^\vee \subset (\mathbb{C}^3)^\vee \otimes (\mathbb{C}^3)^\vee$. It is then a straightforward calculation that $\text{Stab}_G(\varphi)$ is a degree 9 extension of the complex special orthogonal group $\text{SO}(3, \mathbb{C})$, where $\text{SO}(3, \mathbb{C})$ acts diagonally on $U' \otimes V \otimes W$ via the isomorphisms $U' \cong V \cong W \cong \mathbb{C}^3$. The degree 9 extension comes from the extra freedom of two sets of third roots of unity. We have $\dim \text{SO}(3, \mathbb{C}) = 3$, so $\dim \text{Stab}_G(\varphi) = 3$. The subvariety Σ_3 is a G -orbit, so $\text{codim}(\Sigma_3 \subset \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)) = 7$ by orbit-stabiliser. $\square$

4.2. G -semistability. The goal of Subsection 4.2 is to prove the following criterion for G -semistability:

Theorem 4.9. *A tensor $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ is G -semistable if and only if at least one of the following holds:*

- (i) *the associated cubic surface is $\text{SL}(U)$ -semistable; or*
- (ii) *the surface S_φ contains no rank at most 1 points.*

Our strategy is the following:

- (a) We start by showing the conditions of Theorem 4.9 are sufficient.
- (b) We then give some necessary conditions for G -semistability using the Hilbert-Mumford criterion.
- (c) We split the points $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ into the five cases of Proposition 4.1. We then show every tensor φ is either semistable by (a) or unstable by (b), proving the sufficient conditions of (a) are actually necessary.

The first sufficient condition for G -semistability comes from the $\text{SL}(U)$ -semistability of cubic surfaces. Recall the $\text{SL}(V) \times \text{SL}(W)$ -invariant and $\text{SL}(U)$ -equivariant determinant map

$$\det : U^\vee \otimes V^\vee \otimes W^\vee \longrightarrow \text{Sym}^3(U^\vee),$$

sending a tensor φ to its determinant cubic polynomial $\det(\varphi)$. We have

Proposition 4.10. *Let $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ be a tensor such that S_φ is an $\text{SL}(U)$ -semistable cubic surface. Then φ is G -semistable.*

Proof. If $[S_\varphi] \in |\mathcal{O}_{\mathbb{P}U}(3)|$ is $\text{SL}(U)$ -semistable then there is a non-constant, $\text{SL}(U)$ -invariant function $f \in \mathbb{C}[\text{Sym}^3 U]_\bullet$ where $f([S_\varphi]) \neq 0$. The pullback $f \circ \det \in \mathbb{C}[U \otimes V \otimes W]_\bullet$ is a non-constant G -invariant function and $(f \circ \det)(\varphi) = f([S_\varphi]) \neq 0$. So φ is $\text{SL}(U)$ -semistable. $\square$

The $\text{SL}(U)$ -semistability of cubic surfaces is well known, for now all we need is the following result of Mumford [Mum77, Subsection 1.14]:

Proposition 4.11. *A cubic surface $[S] \in |\mathcal{O}_{\mathbb{P}U}(3)|$ is:*

- *$\text{SL}(U)$ -stable if and only if S is canonical with at worst $4A_1$ singularities (equivalently, every singularity on S is an A_1 singularity);*
- *$\text{SL}(U)$ -semistable if and only if S is canonical with at worst $4A_1$ or $3A_2$ singularities (equivalently, every singularity on S is either an A_1 or an A_2 singularity).*

The other sufficient condition comes from rank conditions on the tensor φ .

Proposition 4.12. *There is an irreducible, homogeneous, G -invariant, degree 24 polynomial $R_{24} \in \mathbb{C}[U \otimes V \otimes W]_\bullet$ where $R_{24}(\varphi) = 0$ if and only if the associated cubic surface S_φ contains a rank at most 1 point.*

Proof. Consider the incidence variety

$$\tilde{\mathcal{R}} := \{(\varphi, u, V') \mid \varphi(u, V', -) = 0\} \subset \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee) \times \mathbb{P}U \times \text{Gr}(2, V).$$

The variety $\tilde{\mathcal{R}}$ comes equipped with two projections $\pi_1 : \tilde{\mathcal{R}} \rightarrow \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ and $\pi_2 : \tilde{\mathcal{R}} \rightarrow \mathbb{P}U \times \text{Gr}(2, V)$. We let $\mathcal{R} \subset \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ be the image $\pi_1(\tilde{\mathcal{R}})$ of π_1 , the closed points $\varphi \in \mathcal{R}$ are the tensors that contain a rank at most 1 point. It suffices to prove that $\mathcal{R}$ is a G -invariant, degree 24 prime divisor; note that $\mathcal{R}$ is G -invariant by construction.

We first prove the map $\pi_1 : \tilde{\mathcal{R}} \rightarrow \mathcal{R}$ is generically finite. It suffices to exhibit one tensor with exactly one rank 1 point, consider

$$\varphi_U = \begin{pmatrix} u_0 & u_3 & -u_2 \\ -u_3 & 0 & u_1 \\ u_2 & -u_1 & 0 \end{pmatrix}.$$

One can check from the 2×2 minors that $\varphi_U(u)$ is rank at most 1 if and only if $u = [1 : 0 : 0 : 0] \in \mathbb{P}U$.

Now, the projection π_2 is surjective and the fibre above a point $(u, V') \in \mathbb{P}U \times \text{Gr}(2, V')$ is cut out by 6 independent linear equations on $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee) \times \{(u, V')\}$. So $\tilde{\mathcal{R}}$ is codimension 6 inside $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee) \times \mathbb{P}U \times \text{Gr}(2, V')$. Then we have

$$\text{codim}(\mathcal{R}) = \text{codim}(\tilde{\mathcal{R}}) - \dim(\mathbb{P}U \times \text{Gr}(2, V')) = 6 - 5 = 1.$$

Because the projection π_2 is surjective onto an irreducible variety and the fibres of π_2 are all irreducible and equidimensional, then $\tilde{\mathcal{R}}$ is irreducible. So $\pi_1(\tilde{\mathcal{R}}) = \mathcal{R}$ is an irreducible, closed, G -invariant subscheme of codimension 1, i.e. a G -invariant prime divisor.

To compute the degree we consider a general line $L \subset \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$. The line L is parametrised by a morphism $l : \mathbb{P}^1 \rightarrow \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ sending $[s : t] \mapsto s\varphi + t\varphi'$ for two sufficiently general points $\varphi, \varphi' \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$. The degree of $\mathcal{R}$ is the intersection number $L \cdot \mathcal{R}$, which is the intersection number of the Segre variety $\mathbb{P}V^\vee \times \mathbb{P}W^\vee \subset \mathbb{P}(V^\vee \otimes W^\vee)$ with the image of the bilinear map $f : \mathbb{P}^1 \times \mathbb{P}U \rightarrow \mathbb{P}(V^\vee \otimes W^\vee)$ that sends $([s : t], u) \mapsto s\varphi_U(u) + t\varphi'_U(u)$.

We compute this intersection number in the Chow ring. Let h_1, h_2, h_3 be the respective generators of the Chow rings $A_*(\mathbb{P}^1)$ and $A_*(\mathbb{P}U)$ and $A_*(\mathbb{P}(V^\vee \otimes W^\vee))$. We have $f^*(h_3) = h_1 + h_2$ and $f^*(\mathbb{P}V^\vee \times \mathbb{P}W^\vee) = f^*(6h_3^4) = 6f^*(h_3)^4$. And $(h_1 + h_2)^4 = 4h_1h_2^3$, so $\deg(\mathcal{R}) = 4 \cdot 6 = 24$. $\square$

Definition 4.13. As above, we let $\mathcal{R} \subset \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ be the subvariety parametrising tensors $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ whose associated cubic surface S_φ contains a rank at most 1 point. We let $R_{24} \in \mathbb{C}[U \otimes V \otimes W]_\bullet$ be a generator of the principal ideal cutting out $\mathcal{R} \subset \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$.

An immediate consequence of Proposition 4.12 is

Proposition 4.14. *Let $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee) \setminus \mathcal{R}$ be a tensor with no rank at most 1 points. Then φ is G -semistable.*

Proof. The non-constant, G -invariant polynomial $R_{24} \in \mathbb{C}[U \otimes V \otimes W]_\bullet$ does not vanish at φ . $\square$

Our two sufficient conditions for G -semistability are Propositions 4.10 and 4.14. Before applying the Hilbert-Mumford criterion to prove these two conditions are necessary, we introduce some notation. Let B be a $4 \times 3 \times 3$ array of 0s and *s, in the sense that $B = [B_0|B_1|B_2|B_3]$ and each B_i is a 3×3 matrix of 0s and *. A tensor $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ is said to be of *block form* B if there is a basis for U and V and W where $\varphi_U = \sum_{i=0}^3 u_i \varphi_i$ and each $\varphi_i \in \text{Hom}(V, W^\vee) \cong \text{Hom}(\mathbb{C}^3, \mathbb{C}^3)$ has a 0 in every entry that B_i has a 0 (the matrix φ_i may also have 0s where there is an * in B_i).

Note that we choose the ordering so that $\mathrm{SL}(V)$ acts on the columns of B_i and $\mathrm{SL}(W)$ acts on the rows of B_i .

Proposition 4.15. *A tensor $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ is G -unstable if φ is of one of the following block forms:*

$$(u1) \quad \left[\begin{array}{ccc|ccc|ccc} * & * & * & * & * & * & * & * & * \\ * & * & * & * & * & * & * & * & * \\ * & * & * & * & * & * & * & * & * \end{array} \right] \begin{array}{ccc} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}$$

$$(u2) \quad \left[\begin{array}{ccc|ccc|ccc} * & * & * & * & * & * & * & * & * \\ * & * & * & * & * & * & * & * & * \\ * & * & * & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right] \begin{array}{ccc} * & * & * \\ * & * & * \\ 0 & 0 & 0 \end{array}$$

$$(u2)^T \quad \left[\begin{array}{ccc|ccc|ccc} * & * & * & * & * & 0 & * & * & 0 \\ * & * & * & * & * & 0 & * & * & 0 \\ * & * & * & * & * & 0 & * & * & 0 \end{array} \right] \begin{array}{ccc} * & * & 0 \\ * & * & 0 \\ * & * & 0 \end{array}$$

$$(u3) \quad \left[\begin{array}{ccc|ccc|ccc} * & * & * & * & * & * & * & * & * \\ * & * & * & * & 0 & 0 & * & 0 & 0 \\ * & * & * & * & 0 & 0 & * & 0 & 0 \end{array} \right] \begin{array}{ccc} * & * & * \\ * & 0 & 0 \\ * & 0 & 0 \end{array}$$

$$(u4) \quad \left[\begin{array}{ccc|ccc|ccc} * & * & * & * & * & * & * & * & * \\ * & * & * & * & * & * & * & 0 & 0 \\ * & * & * & * & * & * & * & 0 & 0 \end{array} \right] \begin{array}{ccc} * & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}$$

$$(u5) \quad \left[\begin{array}{ccc|ccc|ccc} * & * & * & * & * & * & * & * & * \\ * & * & * & * & * & * & * & * & * \\ * & * & 0 & * & * & 0 & * & 0 & 0 \end{array} \right] \begin{array}{ccc} * & 0 & 0 \\ * & 0 & 0 \\ 0 & 0 & 0 \end{array}$$

$$(u5)^T \quad \left[\begin{array}{ccc|ccc|ccc} * & * & * & * & * & * & * & * & * \\ * & * & * & * & * & * & * & * & * \\ * & * & 0 & * & * & 0 & 0 & 0 & 0 \end{array} \right] \begin{array}{ccc} * & * & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}$$

Proof. We apply the Hilbert-Mumford criterion (see, for example, Subsection 3.1 of [Hos23]). Let $\lambda : \mathbb{G}_m \rightarrow G$ be a one-parameter subgroup that acts diagonally in a basis for U and V and W . The subgroup λ has *weight vector* $\mathrm{wt}(\lambda) = ((a_0, a_1, a_2, a_3), (b_0, b_1, b_2), (c_0, c_1, c_2)) \in \mathbb{Z}^4 \times \mathbb{Z}^3 \times \mathbb{Z}^3$, in the sense that λ acts as

$$\lambda(t) \cdot \sum_{i,j,k} \varphi_{ijk}(u_i \otimes v_j \otimes w_k) = t^{a_i+b_j+c_k} \varphi_{ijk}(u_i \otimes v_j \otimes w_k).$$

The exponent $a_i + b_j + c_k$ is said to be the *weight* with which λ acts on the vector $u_i \otimes v_j \otimes w_k$. The Hilbert-Mumford criterion states that a point $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ is G -unstable if and only if there exists a one-parameter subgroup $\lambda : \mathbb{G}_m \rightarrow G$ that acts with only *positive* weights on φ , in the sense that the coefficient $\varphi_{ijk} = 0$ whenever $a_i + b_j + c_k \leq 0$. While not all one-parameter subgroups $\lambda : \mathbb{G}_m \rightarrow G$ act diagonally, it is enough to only check the action of diagonal subgroups.

To each weight vector $w = ((a_0, a_1, a_2, a_3), (b_0, b_1, b_2), (c_0, c_1, c_2)) \in \mathbb{Z}^4 \times \mathbb{Z}^3 \times \mathbb{Z}^3$ we associate a block form B , where B has a 0 in the ijk^{th} entry whenever $a_i + b_j + c_k \leq 0$. If a tensor φ is of block form B then there is a basis for U and V and W where $\varphi_{ijk} = 0$ whenever $a_i + b_j + c_k \leq 0$. The

one-parameter subgroup $\lambda_w : \mathbb{G}_m \rightarrow G$ that is diagonal in this basis with weight vector $\text{wt}(\lambda_w) = w$ acts with only positive weights on φ . So φ is G -unstable.

To complete the proof we need to exhibit a weight vector associated to each of the above block forms. The pairs of the weight vector w and its associated G -unstable block form B are the following:

B	w
$(u1)$	$((1, 1, 1, -3), (0, 0, 0), (0, 0, 0))$
$(u2)$	$((9, -3, -3, -3), (4, 4, -8), (0, 0, 0))$
$(u2)^T$	$((9, -3, -3, -3), (0, 0, 0), (4, 4, -8))$
$(u3)$	$((9, -3, -3, -3), (8, -4, -4), (8, -4, -4))$
$(u4)$	$((9, 9, -3, -15), (8, -4, -4), (8, -4, -4))$
$(u5)$	$((9, 9, -3, -15), (12, 0, -12), (4, 4, -8))$
$(u5)^T$	$((9, 9, -3, -15), (4, 4, -8), (12, 0, -12))$

□

The conditions for instability in Proposition 4.15 were found with an algorithmic implementation of the Hilbert-Mumford criterion using convex geometry, see Subsection 3.1 of [Hos23] for details on the convex criterion. This algorithm outputs 18 maximal G -unstable block forms, reducing to 12 after taking transposes, which further reduces to the 5 forms above after suitable changes of basis. We provide these 12 forms in Appendix A for completeness. For the purpose of our proofs we also list block forms $(u10)$ and $(u10)^T$ which are subforms of $(u4)$, and block form $(u12)$ which is a subform of both $(u5)$ and $(u5)^T$.

$(u10)$ associated to the weight vector $((3, 3, -3, -3), (0, 0, 0), (4, -2, -2))$:

$$\left[\begin{array}{ccc|ccc|ccc} * & * & * & * & * & * & * & * & * \\ * & * & * & * & * & * & 0 & 0 & 0 \\ * & * & * & * & * & * & 0 & 0 & 0 \end{array} \right]$$

$(u10)^T$ associated to the weight vector $((3, 3, -3, -3), (4, -2, -2), (0, 0, 0))$:

$$\left[\begin{array}{ccc|ccc|ccc} * & * & * & * & * & * & * & 0 & 0 \\ * & * & * & * & * & * & * & 0 & 0 \\ * & * & * & * & * & * & * & 0 & 0 \end{array} \right]$$

$(u12)$ associated to the weight vector $(3, 3, -3, -3), (2, 2, -4), (2, 2, -4))$:

$$\left[\begin{array}{ccc|ccc|ccc} * & * & * & * & * & * & * & * & 0 \\ * & * & * & * & * & * & * & * & 0 \\ * & * & 0 & * & * & 0 & 0 & 0 & 0 \end{array} \right]$$

In principle, the Hilbert-Mumford criterion also gives a sufficient condition for G -semistability, as long as one can prove they have generated all possible G -unstable block forms. We instead use the sufficient conditions of Propositions 4.10 and 4.14.

The first three G -unstable block forms in Proposition 4.15 have straightforward geometric interpretations. It is easy to check that

Proposition 4.16. *A tensor $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ is G -unstable of block form:*

- $(u1)$ if and only if the map $\varphi_U : U \rightarrow V^\vee \otimes W^\vee$ is not injective;
- $(u2)$ if and only if the surface $S_\varphi \subset \mathbb{P}U$ contains a plane of type $(3, 1)$;

- $(u2)^T$ if and only if the surface $S_\varphi \subset \mathbb{P}U$ contains a plane of type $(1, 3)$;
- $(u3)$ if and only if the surface $S_\varphi \subset \mathbb{P}U$ contains a plane of type $(2, 2)$.

In particular, if a tensor $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ is of one of the block forms $(u1)$ or $(u2)$ or $(u2)^T$ or $(u3)$ then the associated surface S_φ is either a cone or reducible, and conversely every G -unstable determinantal representation of a canonical cubic surface is of block form $(u4)$ or $(u5)$ or $(u5)^T$.

We will now prove that every tensor $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ either satisfies one of the conditions for G -semistability in Propositions 4.10 and 4.14, or is G -unstable by Proposition 4.15. We will tackle each of the five cases of Proposition 4.1 in order, starting with the case that φ is a canonical tensor.

The set of canonical tensors $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ whose associated cubic surface S_φ is $\mathrm{SL}(U)$ -unstable are all of type $F(n)$, where $A_3 \subset F \subsetneq \tilde{E}_6$ is a Dynkin diagram such that $F \neq E_6$. We have the following closure relations between these strata:

Proposition 4.17. *The sets $\overline{A_3(a)}$, $\overline{A_3(b)}$, $\overline{A_3(c)}$ $\subset \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ contain the following subsets:*

$F(n)$	$F'(n')$ appears if and only if $F'(n') \subset \overline{F(n)}$
$A_3(a)$	$A_1A_3(a), A_4(a), A_4(b), 2A_1A_3(a), A_1A_4(a), A_1A_4(b), A_1A_5(a), D_4(a), D_4(b), D_5(a), D_5(b)$
$A_3(b)$	$A_1A_3(d), A_4(a), A_4(c), 2A_1A_3(b), A_1A_4(a), A_1A_4(d), A_5(a), A_1A_5(a), D_4(a), D_4(b), D_5(a), D_5(b)$
$A_3(c)$	$A_1A_3(b), A_1A_3(e), A_4(b), A_4(d), 2A_1A_3(c), A_1A_4(b), A_1A_4(c), A_5(a), A_1A_5(a), D_4(a), D_4(b), D_5(a)$

If $A_3 \subset F \subset \tilde{E}_6$ and $F \neq E_6$ then either $F(n) \subset \overline{A_3(a)} \cup \overline{A_3(b)} \cup \overline{A_3(c)}$ or $F(n) = A_3(d)$ or $F(n) = A_1A_3(f)$.

Proof. The proof amounts to carefully checking all of the closure relations between subschemes of the form $\chi(F(n)) \subset \mathrm{Hilb}_6(\mathbb{P}V)$ using the lists of associated subschemes in [Ng02], and then passing to subschemes of the form $F(n) \subset \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ using Proposition 4.7. $\square$

We are now ready to prove

Lemma 4.18. *Let $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ be a tensor whose associated cubic surface S_φ is canonical. Then φ is G -semistable if and only if either S_φ is $\mathrm{SL}(U)$ -semistable or S_φ contains no rank at most 1 points. A G -semistable tensor φ satisfies exactly one of the following conditions:*

- (i) S_φ has at worst $4A_1$ or $3A_2$ singularities; or
- (ii) $\varphi \in A_3(d)$ and φ_U is G -equivalent to the matrix

$$\begin{pmatrix} (\alpha - 1)u_0 + u_1 & -u_1 & u_0 \\ u_2 & 0 & u_3 \\ u_0 + u_3 & \alpha u_0 + u_1 + u_2 & 0 \end{pmatrix},$$

where $\alpha \in \mathbb{C} \setminus \{0\}$ is a parameter that determines the isomorphism class of S_φ ; or

- (iii) $\varphi \in A_1A_3(f)$ and φ_U is G -equivalent to the matrix

$$\begin{pmatrix} u_1 & u_0 & 0 \\ u_2 & 0 & u_3 - u_0 - u_1 \\ u_3 & u_2 & u_1 \end{pmatrix}.$$

Proof. Note that by Proposition 4.11 (a result of Mumford), a cubic surface S is $\mathrm{SL}(U)$ -semistable if and only if S has at worst $4A_1$ or $3A_2$ singularities. The complement of this condition is that a canonical surface S is $\mathrm{SL}(U)$ -unstable if and only if S has an A_3 (or worse) singularity, in particular $|\mathcal{O}_{\mathbb{P}U}(3)|^{\mathrm{SL}(U)-us} = \overline{\mathcal{S}(A_3)}$.

Fix canonical tensor φ . By Proposition 4.10, if the associated cubic surface S_φ is $\mathrm{SL}(U)$ -semistable then φ is G -semistable. And if $\varphi \in A_3(d)$ or $\varphi \in A_1A_3(f)$ then S_φ has no rank 1 points (this is easy to check from non-vanishing of the 2×2 minors), so φ is G -semistable by Proposition 4.14. By Proposition 4.17, the only remaining possibility is that $\varphi \in \overline{A_3(a)} \cup \overline{A_3(b)} \cup \overline{A_3(c)}$.

So it suffices to prove that every tensor of type $A_3(a)$ or $A_3(b)$ or $A_3(c)$ is G -unstable. Using the classification in [Ng02], this is immediate:

- Every tensor $\varphi \in A_3(a)$ is of the form

$$\varphi = \left[\begin{array}{ccc|ccc|ccc} 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & -1 & 0 & -\alpha & 0 & 0 & 0 & 1 & 0 \end{array} \right]$$

for some $\alpha \in \mathbb{C} \setminus \{0\}$. After permuting some rows and columns, φ is of block form (u4) (the order of the rows, columns, and 3×3 blocks all need to be reversed, the reader can easily apply this permutation by rotating the page by 180 degrees).

- Every tensor $\varphi \in A_3(b)$ is of the form

$$\varphi = \left[\begin{array}{ccc|ccc|ccc} 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & -\beta & 0 & 0 & -\beta & 0 & 1 & -\beta & 0 \end{array} \right]$$

for some $\beta \in \mathbb{C} \setminus \{0\}$. After the same 180 degree rotation, φ is of block form (u5).

- Every tensor $\varphi \in A_3(c)$ is of the form

$$\varphi = \left[\begin{array}{ccc|ccc|ccc} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 1 & 0 & 2\beta - 1 & 0 & 0 & 0 & \beta & 0 & 0 \end{array} \right]$$

for some $\beta \in \mathbb{C} \setminus \{0\}$. After the same 180 degree rotation, φ is of block form $(u5)^T$.

The matrix presentations of types $A_3(d)$ and $A_1A_3(f)$ also come from [Ng02]. $\square$

A general tensor of block form (u4) or (u5) or $(u5)^T$ is of type $A_3(a)$ or $A_3(b)$ or $A_3(c)$, respectively. As a result,

Proposition 4.19. *A tensor $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ is G -unstable of block form:*

- (u4) if and only if $\varphi \in \overline{A_3(a)}$;
- (u5) if and only if $\varphi \in \overline{A_3(b)}$;
- $(u5)^T$ if and only if $\varphi \in \overline{A_3(c)}$.

Propositions 4.16 and 4.19 combine to give a geometric interpretation of the maximal G -unstable block forms that arose from the Hilbert-Mumford criterion, which are listed in Proposition 4.15. We now tackle the case where the surface S_φ is reducible.

Lemma 4.20. *Let $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ be a tensor whose associated cubic surface S_φ is reducible. Then φ is G -semistable if and only if S_φ contains no rank at most 1 points. In this case, φ is a rank 3 skew-symmetric tensor, which is G -equivalent to*

$$\sigma = \begin{pmatrix} u_0 & u_3 & -u_2 \\ -u_3 & u_0 & u_1 \\ u_2 & -u_1 & u_0 \end{pmatrix}.$$

Proof. Because S_φ is reducible then S_φ contains a hyperplane $\mathbb{P}U' \subset \mathbb{P}U$. If $\mathbb{P}U'$ is a plane of type (3, 1) or (1, 3) or (2, 2) then, by Proposition 4.15, the tensor φ is G -unstable of block form $(u2)$ or $(u2)^T$ or $(u3)$, respectively. By Proposition 2.18 the only remaining possibility is that $\varphi_U(U') \subset V^\vee \otimes W^\vee$ is a skew-symmetric subspace.

If φ is a skew-symmetric tensor then, by Proposition 2.20, φ is G -equivalent to the tensor

$$\sigma' = \begin{pmatrix} \lambda_1 u_0 & u_3 & -u_2 \\ -u_3 & \lambda_2 u_0 & u_1 \\ u_2 & -u_1 & \lambda_3 u_0 \end{pmatrix},$$

where each λ_i is either 0 or 1 and $\lambda_1 \geq \lambda_2 \geq \lambda_3$. If $\lambda_3 = 0$ then σ' (and hence φ) is of block form $(u12)$ and is G -unstable. Otherwise, φ is G -equivalent to σ . It is easy to check from the non-vanishing of the 2×2 minors that S_σ has no rank at most 1 points, so every tensor $\varphi \in \Sigma_3 = G \cdot \sigma$ is G -semistable by Proposition 4.14. $\square$

We now tackle the case where the surface S_φ is a cone over a cubic curve.

Lemma 4.21. *Let $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ be a tensor whose associated cubic surface S_φ is a cone over a cubic curve. Then φ is G -unstable.*

Proof. We choose coordinates so that the polynomial $\det(\varphi_U)$ is a polynomial only in the variables u_0, u_1, u_2 . We will split this proof into the four possible ranks of the matrix $\varphi_U(0, 0, 0, 1) \in V^\vee \otimes W^\vee$.

- Suppose $\varphi_U(0, 0, 0, 1)$ is rank 0. Then φ_U is not injective, so φ is G -unstable of block form $(u1)$ by Proposition 4.15.
- Otherwise, suppose $\varphi_U(0, 0, 0, 1)$ is rank 1. Then, up to $\mathrm{SL}(V) \times \mathrm{SL}(W)$ -equivalence, we have

$$\varphi_U = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} + \begin{pmatrix} u_3 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

where the a_{ij} are linear forms in u_0, u_1, u_2 . Computing the determinant, we have

$$\det(\varphi_U) = (a_{22}a_{33} - a_{23}a_{32})u_3 + (\text{terms in } u_0, u_1, u_2).$$

By assumption, the polynomial $\det(\varphi_U)$ has no u_3 term so $a_{22}a_{33} - a_{23}a_{32} = 0$ as polynomials. We can then choose coordinates so that $a_{22}, a_{23}, a_{32}, a_{33}$ are each scalar multiples of either u_0 or u_1 . Then φ is of block form $(u4)$ and is G -unstable.

- Otherwise, suppose $\varphi_U(0, 0, 0, 1)$ is rank 2. Then, up to $\mathrm{SL}(V) \times \mathrm{SL}(W)$ -equivalence, we have

$$\varphi_U = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} + \begin{pmatrix} u_3 & 0 & 0 \\ 0 & u_3 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

where the a_{ij} are linear forms in u_0, u_1, u_2 . We have

$$\det(\varphi_U) = a_{33}u_3^2 + (a_{11}a_{33} + a_{22}a_{33} - a_{13}a_{31} - a_{23}a_{32})u_3 + (\text{terms in } u_0, u_1, u_2).$$

We have $a_{33} = 0$ because the polynomial $\det(\varphi_U)$ has no u_3^2 term. And $a_{13}a_{31} + a_{23}a_{32} = 0$ as polynomials because $\det(\varphi_U)$ has no u_3 term. We can then choose coordinates so that $a_{13}, a_{31}, a_{23}, a_{32}$ are each scalar multiples of either u_0 or u_1 . Then φ is of block form $(u12)$ and is G -unstable.

- It is not possible for $\varphi_U(0, 0, 0, 1)$ to be rank 3: in this case $\det(\varphi_U)$ would contain a non-zero u_3^3 term (whose coefficient is the non-zero determinant of $\varphi_U(0, 0, 0, 1)$).

□

We now tackle the case where the surface S_φ is non-normal, not a cone, and irreducible.

Lemma 4.22. *Let $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ be a tensor whose associated cubic surface S_φ is non-normal, non-conical, and irreducible. Then φ is G -unstable.*

Proof. By Proposition 4.1, the tensor φ is $\mathrm{SL}(U)$ -equivalent to a tensor where either $\det(\varphi_U) = u_2u_0^2 + u_3u_1^2$ or $\det(\varphi_U) = u_2u_0^2 + u_3u_0u_1 + u_1^3$. Either way, we can choose coordinates so that $\det(\varphi_U)$ is at most linear in the variables u_2 and u_3 . We let $L \subset S_\varphi \subset \mathbb{P}U$ be the line parametrised by $[0 : 0 : u_2 : u_3]$.

We first consider the case where the line L is a rank at most 1 linear subspace of φ_U , i.e. that $\varphi_U(0, 0, u_2, u_3)$ is rank 1 for all $[u_2 : u_3]$. By Proposition 2.15, L satisfies a block rank condition of type $(3, 2)$ or $(2, 3)$. But then φ is G -unstable of block form $(u10)$ or $(u10)^T$, respectively.

Otherwise, a general point of L is rank 2. We choose two rank 2 points $l_1, l_2 \in L$ and change coordinates so that $l_1 = [0 : 0 : 1 : 0]$ and $l_2 = [0 : 0 : 0 : 1]$. We let

$$\begin{aligned} V_2 &= \mathrm{coker}(\varphi_U(0, 0, 1, 0)) \subset V, & V_3 &= \mathrm{coker}(\varphi_U(0, 0, 0, 1)) \subset V, \\ W_2 &= \ker(\varphi_U(0, 0, 1, 0)) \subset W, & W_3 &= \ker(\varphi_U(0, 0, 0, 1)) \subset W \end{aligned}$$

be their left and right kernels, which are all 1-dimensional by assumption. We can choose coordinates so that

$$(9) \quad \varphi_U = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} + \begin{pmatrix} u_2 & 0 & 0 \\ 0 & u_2 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

where the a_{ij} are linear forms in u_0, u_1, u_3 . The coefficient of u_2^2 in $\det(\varphi_U)$ is a_{33} , which must be zero by assumption, so $\varphi_{V \otimes W}(V_2 \otimes W_2) = 0$. Similarly, $\varphi_{V \otimes W}(V_3 \otimes W_3) = 0$. Now, we split the rest of the proof into the following four cases:

- Suppose $V_2 = V_3$ and $W_2 = W_3$. Then, using the coordinates of (9), the forms $a_{13}, a_{23}, a_{31}, a_{32}$ also contain no u_3 term and $a_{33} = 0$. So φ is G -unstable of form $(u12)$.
- Otherwise, suppose that $V_2 = V_3$ but $W_2 \neq W_3$. We can choose coordinates so that

$$\varphi_U = \begin{pmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & 0 & 0 \end{pmatrix} + \begin{pmatrix} u_2 & 0 & 0 \\ 0 & u_2 & 0 \\ 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} c_{11} & 0 & c_{13} \\ c_{21} & 0 & c_{23} \\ 0 & 0 & 0 \end{pmatrix},$$

where the b_{ij} are linear forms in u_0, u_1 and the c_{ij} are scalar multiples of u_3 . But then the surface S_φ contains the hyperplane cut out by $b_{31} = 0$ and is reducible, a contradiction (and φ would be G -unstable of block form $(u2)$).

- Similarly, if $W_2 = W_3$ but $V_2 \neq V_3$ then the surface S_φ is reducible.
- Finally, suppose that both $V_2 \neq V_3$ and $W_2 \neq W_3$. We can choose coordinates so that

$$\varphi_U = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} + \begin{pmatrix} \beta_{11} & \beta_{12} & 0 \\ \beta_{21} & \beta_{22} & 0 \\ 0 & 0 & 0 \end{pmatrix} u_2 + \begin{pmatrix} \gamma_{11} & 0 & \gamma_{13} \\ 0 & 0 & 0 \\ \gamma_{31} & 0 & \gamma_{33} \end{pmatrix} u_3,$$

where the a_{ij} are linear forms in u_0, u_1 and the β_{ij} and γ_{ij} are scalars. We have

$$\begin{aligned} \det(\varphi_U) &= (\beta_{11}\beta_{22} - \beta_{12}\beta_{21})(\gamma_{33}u_3 + a_{33})u_2^2 + (\gamma_{11}\gamma_{33} - \gamma_{13}\gamma_{31})(\beta_{22}u_2 + a_{33})u_3^2 \\ &\quad + (\beta_{12}\gamma_{31}a_{23} + \beta_{21}\gamma_{13}a_{32})u_2u_3 + (\text{terms at most linear in } u_2 \text{ and } u_3) \end{aligned}$$

By assumption, the determinant $\det(\varphi_U)$ is at most linear in u_2 and u_3 . Additionally, we assumed the matrices $\varphi_U(0, 0, 1, 0)$ and $\varphi_U(0, 0, 0, 1)$ were rank 2, so $\beta_{11}\beta_{22} - \beta_{12}\beta_{21} \neq 0$ and $\gamma_{11}\gamma_{33} - \gamma_{13}\gamma_{31} \neq 0$. Then $a_{33} = 0$ and $\gamma_{33} = 0$ and $\beta_{22} = 0$.

Because $\det(\varphi_U)$ contains no u_2u_3 term then we also have $\beta_{12}\gamma_{31}a_{23} + \beta_{21}\gamma_{13}a_{32} = 0$ as polynomials. Because the matrices $\varphi_U(0, 0, 1, 0)$ and $\varphi_U(0, 0, 0, 1)$ are each rank 2 then $\beta_{12}\beta_{21} \neq 0$ and $\gamma_{13}\gamma_{31} \neq 0$. So a_{23} and a_{32} are scalar multiples of each other. Choosing coordinates so that $a_{23} = \lambda u_0$ and $a_{32} = \lambda' u_0$ (where λ and λ' may be zero), we have

$$\varphi_U = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & 0 & \lambda u_0 \\ a_{31} & \lambda' u_0 & 0 \end{pmatrix} + \begin{pmatrix} \beta_{11} & \beta_{12} & 0 \\ \beta_{21} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} u_2 + \begin{pmatrix} \gamma_{11} & 0 & \gamma_{13} \\ 0 & 0 & 0 \\ \gamma_{31} & 0 & 0 \end{pmatrix} u_3$$

and S_φ is reducible, a contradiction (and φ would also be G -unstable of block form (u3)). $\square$

Finally, we tackle the case where the determinant cubic vanishes everywhere, i.e. when $S_\varphi = \mathbb{P}U$.

Lemma 4.23. *Let $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ be a tensor such that $S_\varphi = \mathbb{P}U$. Then φ is G -unstable.*

Proof. If $S_\varphi = \mathbb{P}U$ then $\varphi_U(U) \subset V^\vee \otimes W^\vee$ is a rank at most 2 linear subspace. Either the map φ_U is not injective, in which case φ is G -unstable of block form (u1). Or $\varphi_U(U)$ is a 4-dimensional rank at most 2 linear subspace which, by Proposition 2.18, satisfies a block rank condition of type (1, 3) or (3, 1) or (2, 2). Then φ is of block form (u2) or $(u2)^T$ or (u3), respectively, and is G -unstable. $\square$

We can now complete the proof of Theorem 4.9: that a tensor $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ is G -semistable if and only if the surface S_φ is $\mathrm{SL}(U)$ -semistable or S_φ contains no rank at most 1 points. Equivalently, a tensor $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ is G -semistable if and only if either the associated cubic surface S_φ has at worst $4A_1$ or $3A_2$ singularities or $\varphi \in A_3(d) \cup A_1A_3(f) \cup \Sigma_3$.

Proof of Theorem 4.9. By Proposition 4.1, each closed point $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ satisfies (at least) one of five conditions. These five cases are covered in order in Lemmas 4.18, 4.20, 4.21, 4.22 and 4.23. Combining these five lemmas completes the proof. $\square$

The criteria of Theorem 4.9 allow us to describe the G -invariant subring of $\mathbb{C}[U \otimes V \otimes W]_\bullet$ up to finite extension. We will use this description to understand some of the topology of the quotient $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)/G$ and determine the G -stable and G -polystable loci in Subsections 4.3 and 4.4. The $\mathrm{SL}(U)$ -invariant ring $\mathbb{C}[\mathrm{Sym}^3(U)]_\bullet^{\mathrm{SL}(U)}$ of cubic surfaces was known to Clebsch and Salmon in 1860.

Proposition 4.24 ([Sal60; Cle61b; Cle61a]). *The subring of $\mathrm{SL}(U)$ -invariants of $\mathbb{C}[\mathrm{Sym}^3(U)]_\bullet$ is generated by six invariants $I_8, I_{16}, I_{24}, I_{32}, I_{40}, I_{100}$ in degrees 8, 16, 24, 32, 40, 100. The first five invariants are algebraically independent, and I_{100}^2 is a polynomial in the first five invariants.*

As a result, the GIT quotient $|\mathcal{O}_{\mathbb{P}U}(3)|/\mathrm{SL}(U)$ is isomorphic to $\mathbb{P}(1, 2, 3, 4, 5)$. We also note that the discriminant $\Delta \in \mathbb{C}[U \otimes V \otimes W]_\bullet^G$ is a prime divisor of degree 32, where $\Delta([S]) = 0$ if and only if $S \subset \mathbb{P}U$ is a singular surface. We have

$$\Delta = I_{32} + (\text{a polynomial in } I_8, I_{16}, I_{24}).$$

Recall Proposition 4.12, that there is a degree 24 polynomial $R_{24} \in \mathbb{C}[U \otimes V \otimes W]_\bullet$ such that $R_{24}(\varphi) = 0$ if and only if the associated surface S_φ contains a rank at most 1 point.

Definition 4.25. For $d = 8, 16, 24, 32, 40, 100$ we let $J_{3d} := I_d \circ \det$. We define the ring

$$(10) \quad A_\bullet := \mathbb{C}[R_{24}, J_{24}, J_{48}, J_{72}, J_{96}, J_{120}, J_{300}]_\bullet \subset \mathbb{C}[U \otimes V \otimes W]_\bullet^G.$$

We let

$$\eta : \text{Proj}(A_\bullet) \dashrightarrow \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee) // G$$

be the candidate rational map of projective spectra induced by the inclusion (10) of graded rings.

The inclusion of rings $\mathbb{C}[J_{24}, J_{48}, J_{72}, J_{96}, J_{120}, J_{300}]_\bullet \subset A_\bullet$, together with the isomorphism

$$\det^* : \mathbb{C}[\text{Sym}^3(U)]_\bullet^{\text{SL}(U)} = \mathbb{C}[I_8, I_{16}, I_{24}, I_{32}, I_{40}, I_{100}]_\bullet \xrightarrow{\sim} \mathbb{C}[J_{24}, J_{48}, J_{72}, J_{96}, J_{120}, J_{300}]_\bullet,$$

induces a rational map $\xi : \text{Proj}(A_\bullet) \dashrightarrow |\mathcal{O}_{\mathbb{P}U}(3)| // \text{SL}(U)$ and the following diagram commutes:

$$\begin{array}{ccc} \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee) // G & \xrightarrow{\eta} & \text{Proj}(A_\bullet) \\ \searrow \det_{UVW} & & \downarrow \xi \\ & & |\mathcal{O}_{\mathbb{P}U}(3)| // \text{SL}(U). \end{array}$$

We will view $\text{Proj}(A_\bullet)$ as a closed subscheme of the weighted projective space $\mathbb{P}(1, 1, 2, 3, 4, 5)$ using the coordinates $[R_{24} : J_{24} : J_{48} : J_{72} : J_{96} : J_{120}]$. The subscheme $\text{Proj}(A_\bullet)$ is cut out by the relation between R_{24} and the J_{3d} s. In these coordinates, the map ξ is (the restriction of) the forgetful map $\mathbb{P}(1, 1, 2, 3, 4, 5) \dashrightarrow \mathbb{P}(1, 2, 3, 4, 5)$ where we forget the first coordinate R_{24} .

By Theorem 4.9, the ideal generated by the subring $A_\bullet \subset \mathbb{C}[U \otimes V \otimes W]_\bullet$ set-theoretically cuts out the G -unstable locus. As a result, $A_\bullet$ generates the ring of G -invariant functions up to radical and $\mathbb{C}[U \otimes V \otimes W]_\bullet^G$ is a finite, integral extension of its subring $A_\bullet$. In particular, the map η is a finite morphism.

4.3. G -stability. The goal of Subsection 4.3 is to prove a simple criterion for G -stability: that a tensor $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ is G -stable if and only if the associated cubic surface S_φ is $\text{SL}(U)$ -stable. The proof follows a similar structure to that of Theorem 4.9. We start by showing this condition is sufficient and then use the Hilbert-Mumford criterion to prove it is necessary.

Proposition 4.26. *Let $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ be a tensor whose associated cubic surface S_φ is $\text{SL}(U)$ -stable. Then φ is G -stable.*

Proof. Recall that a cubic surface $S \subset \mathbb{P}U$ is $\text{SL}(U)$ -stable if and only if S has at worst $4A_1$ singularities. By Proposition 4.8, if $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ is a tensor whose associated cubic surface S_φ has at worst $4A_1$ singularities then the G -stabiliser $\text{Stab}_G(\varphi)$ is finite. So it suffices to prove that the orbit $G \cdot \varphi \subset \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)^{G-ss}$ is closed inside the G -semistable locus whenever the associated surface S_φ is $\text{SL}(U)$ -stable. We recall the commutative diagram of GIT quotients

$$\begin{array}{ccc} \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee) & \xrightarrow{\det} & |\mathcal{O}_{\mathbb{P}U}(3)| \\ \pi_{UVW} \downarrow & & \downarrow p_U \\ \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee) // G & \xrightarrow{\det_{UVW}} & |\mathcal{O}_{\mathbb{P}U}(3)| // \text{SL}(U). \end{array}$$

Let $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ be a tensor whose associated cubic surface S_φ is $\text{SL}(U)$ -stable and let $\varphi' \in \overline{G \cdot \varphi}^{G-ss}$ be any G -semistable tensor inside its orbit closure. We have $\pi_{UVW}(\varphi) = \pi_{UVW}(\varphi')$. The map $\det_{UVW}$ is defined at $\pi_{UVW}(\varphi)$ because S_φ is $\text{SL}(U)$ -semistable, so we have

$(\det_{UVW} \circ \pi_{UVW})(\varphi) = (\det_{UVW} \circ \pi_{UVW})(\varphi')$. Then $p_U(S_\varphi) = p_U(S_{\varphi'})$ and, because S_φ is $\mathrm{SL}(U)$ -stable, the surfaces S_φ and $S_{\varphi'}$ are $\mathrm{SL}(U)$ -equivalent (because $|\mathcal{O}_{\mathbb{P}^U}(3)|^{\mathrm{SL}(U)-s} // \mathrm{SL}(U)$ is an orbit space).

The G -orbits $G \cdot \varphi$ and $G \cdot \varphi'$ are then irreducible codimension 4 subvarieties of $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$. Since $G \cdot \varphi' \subset \overline{G \cdot \varphi}$, we must have $G \cdot \varphi' = G \cdot \varphi$ and $\varphi' \in G \cdot \varphi$. So the G -orbit $G \cdot \varphi$ is closed inside the G -semistable locus and φ is G -stable whenever its associated cubic surface S_φ is $\mathrm{SL}(U)$ -stable. $\square$

Conversely, by the Hilbert-Mumford criterion we have the necessary condition of

Proposition 4.27. *A tensor $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ is strictly G -semistable if both φ is G -semistable and φ is of one of the following block forms:*

$$\begin{aligned}
 (ns1) \quad & \left[\begin{array}{ccc|ccc|ccc} * & * & * & * & * & * & * & * & * \\ * & * & * & * & * & * & * & * & * \\ * & * & * & * & 0 & 0 & 0 & 0 & 0 \end{array} \right] \\
 (ns1)^T \quad & \left[\begin{array}{ccc|ccc|ccc} * & * & * & * & * & * & * & * & 0 \\ * & * & * & * & * & 0 & * & * & 0 \\ * & * & * & * & * & 0 & * & * & 0 \end{array} \right] \\
 (ns2) \quad & \left[\begin{array}{ccc|ccc|ccc} * & * & * & * & * & * & * & * & 0 \\ * & * & * & * & * & 0 & * & * & 0 \\ * & * & 0 & * & 0 & 0 & * & 0 & 0 \end{array} \right]
 \end{aligned}$$

Proof. We apply the Hilbert-Mumford criterion for stability. A point is not G -stable if and only if there exists a one-parameter subgroup λ that acts with *non-negative* weights on a tensor $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$, i.e. that $\varphi_{ijk} = 0$ whenever $a_i + b_j + c_k < 0$ (where we write $\mathrm{wt}(\lambda) = ((a_0, a_1, a_2, a_3), (b_0, b_1, b_2), (c_0, c_1, c_2))$ for the weight vector). This is almost identical to the Hilbert-Mumford criterion for semistability, the only difference is that we have replaced “positive” with “non-negative” and “ $\leq$ ” with “ $<$ ”. We refer the reader to the proof of Proposition 4.15 for all details, the proof of these two propositions are identical after making those replacements.

To complete the proof, we need to exhibit a weight vector associated to each of the above block forms. The pairs of the weight vector w and its associated not G -stable block form B are the following:

B	w
$(ns1)$	$((3, 0, 0, -3), (2, -1, -1), (1, 1, -2))$
$(ns1)^T$	$((3, 0, 0, -3), (1, 1, -2), (2, -1, -1))$
$(ns2)$	$((1, 0, 0, -1), (1, 0, -1), (1, 0, -1))$

$\square$

Every G -semistable tensor $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)^{G-ss}$ that does not satisfy the conditions of Proposition 4.26 is either of the form $F(n)$ for some Dynkin subdiagram $F \subset 3A_2$ that contains A_2 , or $\varphi \in A_3(d) \cup A_1A_3(f) \cup \Sigma_3$. Either way, φ is contained in the closure $\overline{A_2(a)} \cup \overline{A_2(b)}$. More precisely, we have

Proposition 4.28. *Fix a stratum $F(n) \subset \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$. Then $F(n) \not\subset \Sigma_3$. We have $\Sigma_3 \subset \overline{F(n)}$ if and only if the associated cubic surface S_φ to every tensor $\varphi \in F(n)$ contains no rank at most 1 points.*

Proof. By Proposition 4.8 we have $\dim F(n) \geq 6$ and $\dim \Sigma_3 = 7$, so $\overline{F(n)} \not\subset \Sigma_3$. To prove the rest of this proposition we use a similar method to the proof of Proposition 4.7.

Fix a stratum $F(n) \subset \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee) \setminus \mathcal{R}$ such that the associated cubic surface S_φ to every tensor $\varphi \in F(n)$ contains no rank at most 1 points. As illustrated in Example 4.4, the subvariety $\chi(F(n)) \subset \text{Hilb}_6(\mathbb{P}V)$ parametrises schemes $[X] \in \text{Hilb}_6(\mathbb{P}V)$ satisfying some number of closed conditions (a) and some number of conditions (b), including the condition that X is not contained in a conic. We let $\mathcal{A} \subset \text{Hilb}_6(\mathbb{P}V)$ be the subscheme cut out by the closed conditions (a) and the open conditions (b), with the modification that $[X] \in \mathcal{A}$ is allowed to be contained in at most one conic. The additional points in $\mathcal{A} \setminus \chi(F(n))$ parametrise a certain subset of the set of canonical subschemes $X \subset \text{Hilb}_6(\mathbb{P}V)$ such that

- (i) X is contained in a unique conic and
- (ii) the associated Dynkin diagram F_X to X is the union of F and a point.

Note that we have $\mathcal{A} \subset \overline{\chi(F(n))}$ by construction.

The contraction $\gamma : \text{Hilb}_6(\mathbb{P}V) \dashrightarrow \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee) // (\text{SL}(U) \times \text{SL}(W))$ is defined on $\mathcal{A}(F(n))$, so

$$(11) \quad \gamma(\mathcal{A}) \subset \overline{\gamma(\chi(F(n)))} = \overline{\pi_{UW}(F(n))}.$$

The subschemes $[X] \in \mathcal{A} \setminus \chi(F(n))$ contain no length 3 collinear subscheme because no tensor $\varphi \in F(n)$ contains a rank at most 1 point. This means that each subscheme X must be contained in a smooth conic, so γ contracts $\mathcal{A} \setminus \chi(F(n))$ onto $\pi_{UW}(\Sigma_3)$ (see Proposition 3.16). Then $\gamma(\mathcal{A}) = \pi_{UW}(F(n)) \cup \pi_{UW}(\Sigma_3)$. Because $F(n) \cup \Sigma_3$ is contained in the $\text{SL}(U) \times \text{SL}(W)$ -stable locus then we can lift (11) to $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ by applying π_{UW}^{-1} to both sides, which gives $\Sigma_3 \cup F(n) \subset \overline{F(n)}$.

Conversely, if there is a tensor $\varphi \in F(n)$ whose associated cubic surface S_φ contains no rank at most 1 points (either all points of $F(n)$ satisfy this, or none of them do) then the function R_{24} vanishes on $F(n)$. So the closure $\overline{F(n)}$ is contained in the closed subscheme $\mathcal{R} \subset \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ of tensors φ whose associated surface S_φ contains a rank at most 1 point. We have $\Sigma_3 \cap \mathcal{R} = \emptyset$, so $\Sigma_3 \not\subset \overline{F(n)}$. $\square$

Proposition 4.29. *If $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)^{G-ss}$ is a G -semistable tensor whose associated cubic surface S_φ is not $\text{SL}(U)$ -stable then $\varphi \in \overline{A_2(a)} \cup \overline{A_2(b)}$.*

Proof. This proof is almost identical to that of Proposition 4.17. The proof amounts to carefully checking all of the closure relations between subschemes of the form $\chi(F(n)) \subset \text{Hilb}_6(\mathbb{P}V)$ using the lists of associated subschemes in [Ng02], and then passing to $F(n) \subset \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ by using Proposition 4.7. And we have $\Sigma_3 \subset \overline{A_2(b)}$ by Proposition 4.28. $\square$

We can now prove the following criterion for G -stability:

Theorem 4.30. *A tensor $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ is G -stable if and only if the associated cubic surface S_φ is $\text{SL}(U)$ -stable.*

Proof. Fix a G -semistable tensor $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)^{G-ss}$. By Proposition 4.29, either $\varphi \in \overline{A_2(a)} \cup \overline{A_2(b)}$ or the surface S_φ is $\text{SL}(U)$ -stable. In the latter case, φ is G -stable by Proposition 4.26. So it suffices to prove that all tensors of types $A_2(a)$ and $A_2(b)$ are strictly G -semistable. We again refer the reader to [Ng02] for the following descriptions of the points of $A_2(a)$ and $A_2(b)$.

- Every tensor $\varphi \in A_2(a)$ is of the form

$$\varphi = \left[\begin{array}{ccc|ccc|ccc|ccc} 0 & 0 & 1 & 0 & 0 & 0 & 0 & -\alpha & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & -1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \lambda & 1 & 0 & 0 & 0 & 0 & 1 \end{array} \right]$$

for some $\alpha \in \mathbb{C} \setminus \{0, 1\}$ and $\lambda \in \mathbb{C} \setminus \{-1, 0\}$. Viewing the tensor as $\varphi = [\varphi_0|\varphi_1|\varphi_2|\varphi_3]$, we change coordinates to $\varphi \sim [\varphi_0|\varphi_1 - \lambda\varphi_3|\varphi_3|\varphi_2]$ and see that φ is of block form $(ns1)$. Then φ is not G -stable by Proposition 4.27. A slightly more complicated change of coordinates shows that φ is also of block form $(ns1)^T$.

- Every tensor $\varphi \in A_2(b)$ is of the form

$$\varphi = \left[\begin{array}{ccc|ccc|ccc|ccc} 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\beta & 0 & 0 & 1 - \beta & 1 & -\lambda & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 - \lambda & 0 & 1 & 0 & 0 \end{array} \right]$$

for some $\lambda \in \mathbb{C} \setminus \{0, 1\}$ and $\beta \in \mathbb{C} \setminus \{0, (1 - \lambda)^{-1}\}$. Viewing $\varphi = [\varphi_0|\varphi_1|\varphi_2|\varphi_3]$, we change coordinates to $\varphi \sim [\varphi_1|\varphi_0|\varphi_3|\varphi_2]$. We then swap the first two columns and permute the rows by $(w_0, w_1, w_2) \mapsto (w_2, w_0, w_1)$ (recalling the rows are indexed by the basis of $W^\vee$). We then have

$$\varphi \sim \left[\begin{array}{ccc|ccc|ccc|ccc} 0 & 0 & 1 - \beta & 0 & 0 & -\beta & 0 & 0 & 1 & -\lambda & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 1 - \lambda & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

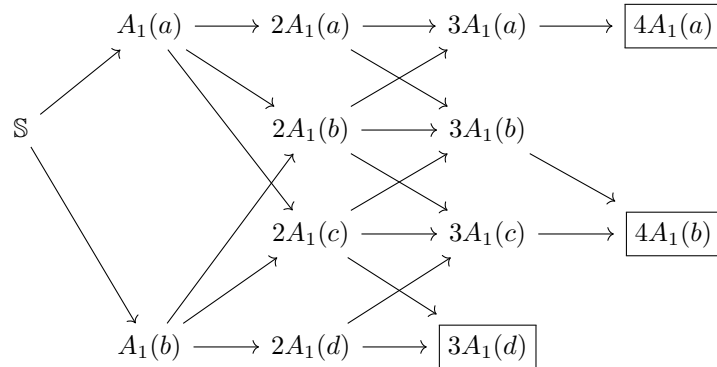
and we see that φ is of block form $(ns2)$ and is not G -stable. □

4.4. Geometry of the GIT quotient. We conclude with a description of the G -polystable locus of $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ and some of the geometry of the GIT quotient $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)/G$. Inside the $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)^{G-ss}$ we have

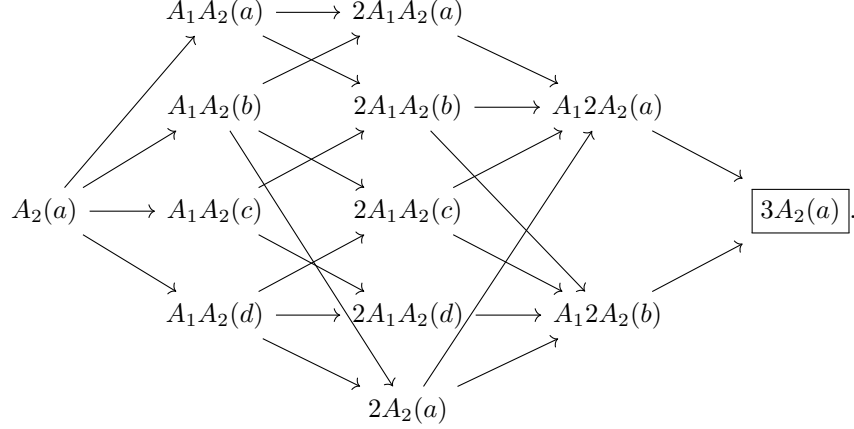
Proposition 4.31. *The G -semistable locus $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)^{G-ss}$ decomposes as the disjoint union*

$$\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)^{G-ss} = \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)^{G-s} \cup \overline{A_2(a)}^{G-ss} \cup \overline{A_2(b)}^{G-ss}.$$

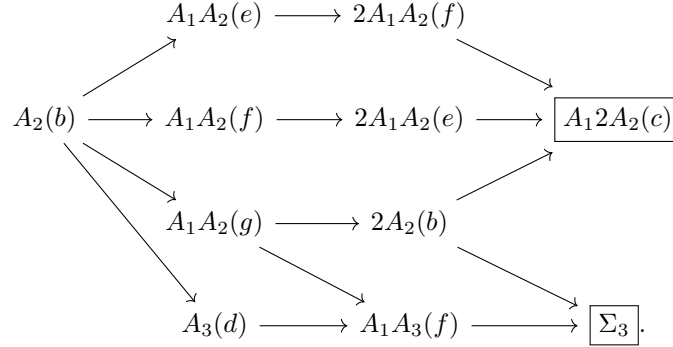
The G -stable locus $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)^{G-s}$ admits the following stratification, where we draw an arrow $Z \rightarrow Z'$ between strata if and only if $Z' \subset \overline{Z}$ (we omit arrows made redundant by transitivity). We draw a box $\boxed{Z}$ to indicate that Z is closed inside $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)^{G-s}$.



Similarly, the codimension 2 subvariety $\overline{A_2(a)}^{G-ss}$ admits the following stratification, where a box $\boxed{Z}$ indicates that Z is closed inside $\overline{A_2(a)}^{G-ss}$:



Finally, the codimension 2 subvariety $\overline{A_2(b)}^{G-ss}$ admits the following stratification, where a box $\boxed{Z}$ indicates that Z is closed inside $\overline{A_2(b)}^{G-ss}$:



Proof. We again apply Propositions 4.7 and 4.28, using the list of associated length 6 subschemes in [Ng02]. $\square$

If φ is a tensor of type $F(n)$ where $|F| \geq 4$ and $F \neq 2A_2$ then $F(n) = G \cdot \varphi$ is a G -orbit. As a result, Proposition 4.31 gives a partial answer to the question of which strictly G -semistable tensors $\varphi' \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)^{G-sss}$ are contained in the orbit closure $\overline{G \cdot \varphi}$ of another strictly G -semistable tensor $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)^{G-sss}$. To properly answer this question, we first recall some standard facts about G -polystability. These can be found in, for example, Sections 4 and 5 of Victoria Hoskins' notes [Hos16].

Definition 4.32. Let T be a vector space and let H be a reductive group acting linearly on $\mathbb{P}T$. A H -semistable point $t \in \mathbb{P}T$ is said to be *H -polystable* if and only if the orbit $H \cdot t$ is closed inside the H -semistable locus $\mathbb{P}T^{H-ss}$. If t is also strictly G -semistable then we say t is *strictly H -polystable*.

We let $\mathbb{P}T^{H-sps} \subset \mathbb{P}T$ be the set of strictly H -polystable points.

Note that all H -stable points are also H -polystable. The subvariety P is closed inside $\mathbb{P}T^{H-ss}$. For every strictly H -semistable point $t \in \mathbb{P}T^{H-sss}$ there is a strictly H -polystable point $s \in P$, unique up to H -equivalence, such that $s \in \overline{H \cdot t}$. Then, under the GIT quotient $p : \mathbb{P}T \dashrightarrow \mathbb{P}T//H$, we have $p(s) = p(t)$. Conversely, for any point $\bar{s} \in \mathbb{P}T//H$ the preimage $p^{-1}(\bar{s}) \subset \mathbb{P}T$ contains a unique closed H -orbit $H \cdot s$ and all other points $t \in p^{-1}(\bar{s})$ satisfy $s \in \overline{H \cdot t}$. As a result, the closed embedding

$$\iota : \mathbb{P}T^{G-ss}/H \longrightarrow \mathbb{P}T^{G-ss}/H$$

is a bijective morphism, although ι may not be an isomorphism of schemes. The $\mathrm{SL}(U)$ -polystability of cubic surfaces is a result of [Mum77, Subsection 1.14].

Proposition 4.33. *A strictly $\mathrm{SL}(U)$ -semistable cubic surface $S \subset \mathbb{P}U$ is $\mathrm{SL}(U)$ -polystable if and only if S has $3A_2$ singularities. The GIT quotient $|\mathcal{O}_{\mathbb{P}U}(3)|^{\mathrm{SL}(U)-sss}/\mathrm{SL}(U)$ of the strictly $\mathrm{SL}(U)$ -semistable locus is a singleton.*

Before stating the corresponding result for $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$, we prove a dimension equality. Recall from Definition 4.13 that $\mathcal{R} \subset \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ is the subvariety parametrising tensors $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ containing a rank at most 1 point; $\mathcal{R} \subset \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ is cut by the principal ideal $(R_{24}) \subset \mathbb{C}[U \otimes V \otimes W]_\bullet$.

Proposition 4.34. *Let $F(n) \subset \mathcal{R}^{G-ss}$ be a stratum where $F \subset 3A_2$ or $F \subset 4A_1$. We have*

$$\dim(\overline{F(n)}/G) = \dim(\overline{\mathcal{S}(F)}/\mathrm{SL}(U)).$$

Proof. We recall the finite, integral extension $A_\bullet \subset \mathbb{C}[U \otimes V \otimes W]_\bullet^G$ from Definition 4.25. This extension remains finite and integral after quotienting both sides by (R_{24}) . By [Sal60], the discriminant $\Delta \in \mathbb{C}[\mathrm{Sym}^3 U]_\bullet^G$ is degree 32 and we have

$$\Delta = I_{32} + \text{a polynomial in } I_8, I_{16}, I_{24}.$$

$$\Delta \circ \det = J_{32} + \text{a polynomial in } J_8, J_{16}, J_{24}.$$

If $\varphi \in \mathcal{R}$ then the surface S_φ is singular, so the polynomial R_{24} divides the pullback $\Delta \circ \det$ of the discriminant Δ . There are no other relations between R_{24} and the J_{3d} s, so we have isomorphisms

$$A_\bullet/(R_{24}) = A_\bullet/(R_{24}, \Delta \circ \det) \cong \mathbb{C}[J_{24}, J_{48}, J_{72}, J_{300}] \cong \mathbb{C}[I_8, I_{16}, I_{24}, I_{100}] = \mathbb{C}[\mathrm{Sym}^3 U]_\bullet^{\mathrm{SL}(U)}/(\Delta).$$

Taking projective spectra, we have

$$\mathcal{R}/G \xrightarrow{\eta|_{\mathcal{R}/G}} \mathrm{Proj}(A_\bullet/(R_{24})) \xrightarrow{\sim} \overline{\mathcal{S}(A_1)}/\mathrm{SL}(U).$$

The map $\eta|_{\mathcal{R}/G}$ is finite. Writing $\eta_{\mathcal{R}} := \eta|_{\mathcal{R}/G}$, the inverse image $\eta_{\mathcal{R}}^{-1}(\overline{\mathcal{S}(F)}/\mathrm{SL}(U))$ decomposes as the disjoint union of the connected components $\overline{F(n)}/G$ (where $F(n) \subset \mathcal{R}$). The restriction of $\eta_{\mathcal{R}}$ to each connected component $\overline{F(n)}/G$ is a finite, surjective morphism onto its image $\overline{\mathcal{S}(F)}/\mathrm{SL}(U)$. The desired equality then follows. $\square$

Theorem 4.35. *A tensor $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ is G -polystable if and only if either $\varphi \in 3A_2(a)$ or $\varphi \in \overline{2A_2(b)}^{G-ss} = 2A_2(b) \cup A_1 2A_2(c) \cup \Sigma_3$.*

The quotient map π_{UVW} (set-theoretically) contracts the stratum $\overline{A_2(a)}^{G-ss}$ onto the singleton $3A_2(a)/G$, and π_{UVW} (set-theoretically) contracts the stratum $\overline{A_2(b)}^{G-ss}$ onto $\overline{2A_2(b)}/G$. There is a bijective morphism $\mathbb{P}^1 \rightarrow \overline{A_2(b)}/G$.

Proof. Recall that $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)^{G-sss}$ decomposes as the disjoint union of its two connected components $\overline{A_2(a)}^{G-ss}$ and $\overline{A_2(b)}^{G-ss}$. We start with the connected component $\overline{A_2(a)}^{G-ss}$. By Proposition 4.31 the subvariety $3A_2(a)$ is closed inside $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)^{G-ss}$ and $3A_2(a)$ is a G -orbit, so $3A_2(a) \subset \overline{A_2(a)}^{G-ss}$. To prove all other points $\varphi \in \overline{A_2(a)}^{G-ss}$ are not G -polystable it suffices to show that $\overline{A_2(a)}/G$ is a singleton. Because $\overline{A_2(a)}^{G-ss}$ is connected then we only need to show that its quotient $\overline{A_2(a)}^{G-ss}/G$ is zero-dimensional. This is an immediate consequence of Propositions 4.33 and 4.34.

The other connected component $\overline{A_2(b)}^{G-ss}$ is more complicated. By [Ng02], a tensor $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ is contained in $\overline{2A_2(b)}^{G-ss}$ if and only if φ is G -equivalent to a tensor of the form

$$\varphi_{[s:t]} = \begin{pmatrix} 0 & u_3 & u_2 \\ u_3 & su_1 + tu_2 & u_0 \\ u_1 & u_0 & 0 \end{pmatrix},$$

where $[s : t] \in \mathbb{P}^1$. If $[s : t] \in \{[1 : 0], [0 : 1]\}$ then $\varphi_{[s:t]} \in A_1 2A_2(c)$. If $\varphi = [1 : 1]$ then $\varphi \in \Sigma_3$. Otherwise, $\varphi_{[s:t]} \in 2A_2(b)$. Two tensors $\varphi_{[s:t]}$ and $\varphi_{[s':t']}$ are G -equivalent if and only if $[s : t] = [s' : t']$ or $[s : t] = [t' : s']$. This equivalence comes from the involution $\tau \in G$ that swaps u_1 and u_2 , v_0 and v_2 , and w_0 and w_2 , which identifies $\tau \cdot \varphi_{[s:t]} = \varphi_{[t:s]}$. Geometrically, τ swaps the two A_2 singularities of S_φ inside $\mathbb{P}U$ and swaps the two length 3 points of X_φ and Y_φ inside $\mathbb{P}V$ and $\mathbb{P}W$, respectively.

We first prove that all tensors $\varphi \in \overline{2A_2(b)}^{G-ss}$ are G -polystable, i.e. that $G \cdot \varphi$ is closed inside $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)^{G-ss}$. By Proposition 4.31, this is immediate if $\varphi \in A_1 2A_2(c) \cup \Sigma_3$. So suppose that $\varphi \in 2A_2(b)$ and let $\varphi' \in \overline{2A_2(b)}^{G-ss}$ be any other tensor. There are two possibilities:

- If $\varphi' \in 2A_2(b) \cup A_1 2A_2(c)$ then, by Proposition 4.8, both φ and φ' have 1-dimensional G -stabilisers and their G -orbits are irreducible of codimension 5. So $G \cdot \varphi' \not\subset \overline{G \cdot \varphi}$ and $\varphi' \notin \overline{G \cdot \varphi}$.
- Otherwise $\varphi' \in \Sigma_3$. In this case, recalling the map η from Definition 4.25, we have $\eta(\pi_{UVW}(\varphi')) = [\lambda : 0 : 0 : 0 : 0 : 0]$ and $\eta(\pi_{UVW}(\varphi)) = [\lambda : 1 : 8 : 0 : 0 : 0]$ for some $\lambda \in \mathbb{C}$ (the cubic invariants I_d all vanish at $[S_\sigma]$ when $\sigma \in \Sigma_3$, but not at $[S_\varphi]$). Then the points φ and φ' are separated by G -invariant functions so $\varphi' \notin \overline{G \cdot \varphi}$.

We now need to prove that φ is not G -polystable whenever $\varphi \in \overline{A_2(b)}^{G-ss} \setminus \overline{2A_2(b)}^{G-ss}$. By Proposition 4.8, the G -stabiliser of every element $\varphi \in \overline{A_2(b)}^{G-ss} \setminus \overline{2A_2(b)}^{G-ss}$ is finite. If a G -semistable point $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ has a finite G -stabiliser then φ is either G -stable (if $G \cdot \varphi$ is closed inside $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$) or φ is strictly G -semistable (otherwise). By assumption, φ is not G -stable, so φ must also not be G -polystable.

The map $f : \mathbb{P}^1 \rightarrow \overline{2A_2(b)}^{G-ss}$ sending $[s : t] \mapsto \varphi_{[s:t]}$ descends to a surjective morphism $(\pi_{UVW} \circ f) : \mathbb{P}^1 \rightarrow \overline{2A_2(b)}/G$. The map $\pi_{UVW} \circ f$ factors through the quotient of $\mathbb{P}^1$ by the subgroup $\{1, \tau\} \subset \text{PGL}(2)$ that acts as $\tau \cdot [s : t] = [t : s]$. The induced map $g : \mathbb{P}^1/\{1, \tau\} \rightarrow \overline{2A_2(b)}/G$ is a bijective morphism, and $\mathbb{P}^1/\{1, \tau\} \cong \mathbb{P}(1, 2) \cong \mathbb{P}^1$. $\square$

The three Theorems 4.9 and 4.30 and 4.35 combine to prove Theorem A. The stratification of $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)^{G-ss}$ by $\mathbb{S}$ and Σ_3 and the strata of type $F(n)$ given in Proposition 4.31 descends to a stratification of the quotient $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)/G$ by $\mathbb{S}/G$ and Σ_3/G and the strata of type $F(n)/G$. We have

Proposition 4.36. *The dimensions of the strata of $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ are the following:*

- $\dim(\mathbb{S}/G) = 4$;
- if $F \subset 4A_1$ then $\dim(F(n)/G) = 4 - |F|$;
- if $A_2 \subset F \subset 3A_2$ then either
 - $F(n) = A_2(b), A_1A_2(g), 2A_2(b)$, in which case $\dim(F(n)/G) = 1$, or
 - $\dim(F(n)/G) = 0$; or
- $\dim(\Sigma_3/G) = 0$.

Proof. The equalities for the G -stable strata follow from using orbit-stabiliser and Proposition 2.4. All other equalities follow from Theorem 4.35. By Proposition 4.31, the strata $A_2(b)$ and $A_1A_2(g)$ are the only strictly G -semistable strata of type $F(n)$ whose closures $\overline{F(n)}$ contain $2A_2(b)$. We then have surjective maps $A_2(b)/G \rightarrow \overline{2A_2(b)}/G$ and $A_1A_2(g)/G \rightarrow (\overline{2A_2(b)} \setminus \Sigma_3)/G$ which are bijections on $\mathbb{C}$ -points, giving the desired equality of dimensions. $\square$

Remark 4.37. The GIT quotient $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)/G$ is essentially the last interesting moduli space of $n \times 3 \times 3$ tensors. Specifically, fix $0 < n < 9$ and let $\varphi_{V \otimes W} : V \otimes W \rightarrow \mathbb{C}^n$ be a surjective map. The kernel map

$$\varphi^\perp : \mathbb{C}^{9-n} \cong \ker(\varphi_{V \otimes W}) \longrightarrow V^\vee \otimes W^\vee$$

is adjoint to a $(9-n) \times 3 \times 3$ tensor. The duality between injective $n \times 3 \times 3$ tensors and injective $(9-n) \times 3 \times 3$ tensors induces the standard isomorphism

$$f : \text{Gr}(n, V^\vee \otimes W^\vee) \xrightarrow{\sim} \text{Gr}(9-n, V \otimes W)$$

of Grassmannians. Because $\text{Gr}(n, V^\vee \otimes W^\vee) \cong \mathbb{P}(\mathbb{C}^n \otimes V^\vee \otimes W^\vee)/(\text{SL}(V) \times \text{SL}(W))$, then f induces an isomorphism of GIT quotients

$$f_{VW} : \mathbb{P}(\mathbb{C}^n \otimes V^\vee \otimes W^\vee)/(\text{SL}(n) \times \text{SL}(V) \times \text{SL}(W)) \xrightarrow{\sim} \mathbb{P}(\mathbb{C}^{9-n} \otimes V \otimes W)/(\text{SL}(n) \times \text{SL}(V) \times \text{SL}(W)).$$

When $n = 1, 2, 7, 8$ there are finitely many $n \times 3 \times 3$ -tensors up to $\text{SL}(n) \times \text{SL}(3) \times \text{SL}(3)$ -equivalence. Ng described the 2-dimensional moduli space for $n = 3$ (and hence $n = 6$) in [Ng95].

The geometry of the associated $5 \times 3 \times 3$ tensor $\varphi^\perp$ to a $4 \times 3 \times 3$ tensor was known to Segre in [Seg87]. When the associated cubic surface S_φ is smooth, the intersection

$$Z_\varphi := \mathbb{P} \ker(\varphi_{V \otimes W}) \cap (\mathbb{P}V \times \mathbb{P}W) \subset \mathbb{P}(V \otimes W)$$

consists of 6 points $Z_\varphi = \{x_1 \otimes y_1, \dots, x_6 \otimes y_6\}$. The projections of Z_φ onto $\mathbb{P}V$ and $\mathbb{P}W$ recover the length 6 subschemes X_φ and Y_φ . The cubic threefold $T_\varphi \subset \mathbb{P} \ker(\varphi_{V \otimes W})$ cut out by $\det(\varphi^\perp) = 0$ has $6A_1$ singularities: one at each point $x_i \otimes y_i \in Z_\varphi$. Conversely, every cubic threefold with $6A_1$ singularities has a unique determinantal representation (up to $\text{GL}(V) \times \text{GL}(W)$ -equivalence and the involution swapping V and W).

APPENDIX A. HILBERT-MUMFORD BLOCK FORMS

All computations were carried out in SageMath version 10.3 [Sage].

G -instability: Our computational implementation of the convex Hilbert-Mumford criterion for G -semistability of $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ outputs the following 12 maximal block forms (and their transposes) for G -instability. A tensor $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ is G -unstable if and only if it is of one of the following block forms:

(u1) associated to the weight vector $((1, 1, 1, -3), (0, 0, 0), (0, 0, 0))$:

$$\left[\begin{array}{ccc|ccc|ccc} * & * & * & * & * & * & * & * & * \\ * & * & * & * & * & * & * & * & * \\ * & * & * & * & * & * & * & * & * \end{array} \right] \begin{array}{ccc} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}$$

(u2) associated to the weight vector $((9, -3, -3, -3), (0, 0, 0), (4, 4, -8))$:

$$\left[\begin{array}{ccc|ccc|ccc} * & * & * & * & * & * & * & * & * \\ * & * & * & * & * & * & * & * & * \\ * & * & * & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right] \begin{array}{ccc} * & * & * \\ * & * & * \\ 0 & 0 & 0 \end{array}$$

(u3) associated to the weight vector $((9, -3, -3, -3), (8, -4, -4), (8, -4, -4))$:

$$\left[\begin{array}{ccc|ccc|ccc} * & * & * & * & * & * & * & * & * \\ * & * & * & * & 0 & 0 & * & 0 & 0 \\ * & * & * & * & 0 & 0 & * & 0 & 0 \end{array} \right] \begin{array}{ccc} * & * & * \\ * & 0 & 0 \\ * & 0 & 0 \end{array}$$

(u4) associated to the weight vector $((9, 9, -3, -15), (8, -4, -4), (8, -4, -4))$:

$$\left[\begin{array}{ccc|ccc|ccc} * & * & * & * & * & * & * & * & * \\ * & * & * & * & * & * & * & 0 & 0 \\ * & * & * & * & * & * & * & 0 & 0 \end{array} \right] \begin{array}{ccc} * & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}$$

(u5) associated to the weight vector $((9, 9, -3, -15), (12, 0, -12), (4, 4, -8))$:

$$\left[\begin{array}{ccc|ccc|ccc} * & * & * & * & * & * & * & * & * \\ * & * & * & * & * & * & * & * & * \\ * & * & 0 & * & * & 0 & * & 0 & 0 \end{array} \right] \begin{array}{ccc} * & 0 & 0 \\ * & 0 & 0 \\ 0 & 0 & 0 \end{array}$$

Contained in form (u2):

(u6) associated to the weight vector $((1, 1, 1, -3), (4, 0, -4), (4, 0, -4))$:

$$\left[\begin{array}{ccc|ccc|ccc} * & * & * & * & * & * & * & * & * \\ * & * & 0 & * & * & 0 & * & * & 0 \\ * & 0 & 0 & * & 0 & 0 & * & 0 & 0 \end{array} \right] \begin{array}{ccc} * & * & 0 \\ * & 0 & 0 \\ 0 & 0 & 0 \end{array}$$

(u7) associated to the weight vector $((12, 0, -12), (16, 4, -20), (9, 9, -3, -15))$:

$$\left[\begin{array}{ccc|ccc|ccc} * & * & * & * & * & * & * & * & * \\ * & * & * & * & * & * & * & * & * \\ * & 0 & 0 & * & 0 & 0 & 0 & 0 & 0 \end{array} \right] \begin{array}{ccc} * & * & 0 \\ * & 0 & 0 \\ 0 & 0 & 0 \end{array}$$

(u8) associated to the weight vector $((3, 3, -3, -3), (4, -2, -2), (6, 0, -6))$:

$$\left[\begin{array}{ccc|ccc|ccc} * & * & * & * & * & * & * & * & * \\ * & * & * & * & * & * & * & 0 & 0 \\ * & 0 & 0 & * & 0 & 0 & 0 & 0 & 0 \end{array} \right] \begin{array}{ccc} * & * & * \\ * & 0 & 0 \\ 0 & 0 & 0 \end{array}$$

(u9) associated to the weight vector $((3, 3, 3, -9), (8, -4, -4), (4, 4, -8))$:

$$\left[\begin{array}{ccc|ccc|ccc} * & * & * & * & * & * & * & * & * \\ * & * & * & * & * & * & * & * & * \\ * & 0 & 0 & * & 0 & 0 & * & 0 & 0 \end{array} \right] \begin{array}{ccc} * & 0 & 0 \\ * & 0 & 0 \\ 0 & 0 & 0 \end{array}$$

Contained in form (u4):

(u10) associated to the weight vector $((3, 3, -3, -3), (0, 0, 0), (4, -2, -2))$:

$$\left[\begin{array}{ccc|ccc|ccc} * & * & * & * & * & * & * & * & * \\ * & * & * & * & * & * & 0 & 0 & 0 \\ * & * & * & * & * & * & 0 & 0 & 0 \end{array} \right]$$

(u11) associated to the weight vector $((5, 1, 1, -7), (4, 0, -4), (4, 0, -4))$:

$$\left[\begin{array}{ccc|ccc|ccc} * & * & * & * & * & * & * & 0 & 0 \\ * & * & * & * & * & 0 & * & 0 & 0 \\ * & * & 0 & * & 0 & 0 & * & 0 & 0 \end{array} \right].$$

Contained in form (u5):

(u12) associated to the weight vector $(3, 3, -3, -3), (2, 2, -4), (2, 2, -4))$:

$$\left[\begin{array}{ccc|ccc|ccc} * & * & * & * & * & * & * & * & 0 \\ * & * & * & * & * & * & * & * & 0 \\ * & * & 0 & * & * & 0 & 0 & 0 & 0 \end{array} \right]$$

Strict G -semistability: Our computational implementation of the convex Hilbert-Mumford criterion for G -(semi)stability of $\mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ outputs the following 7 maximal block forms (and their transposes) for strict G -semistability, in addition to the block forms for G -unstability. A G -semistable tensor $\varphi \in \mathbb{P}(U^\vee \otimes V^\vee \otimes W^\vee)$ is strictly G -semistable if and only if it is of one of the following block forms:

(ns1) associated to the weight $((3, 0, 0, -3), (2, -1, -1), (1, 1, -2))$:

$$\left[\begin{array}{ccc|ccc|ccc} * & * & * & * & * & * & * & 0 & 0 \\ * & * & * & * & * & * & * & 0 & 0 \\ * & * & * & * & 0 & 0 & * & 0 & 0 \end{array} \right]$$

(ns2) associated to the weight $((1, 0, 0, -1), (1, 0, -1), (1, 0, -1))$:

$$\left[\begin{array}{ccc|ccc|ccc} * & * & * & * & * & * & * & * & 0 \\ * & * & * & * & * & 0 & * & 0 & 0 \\ * & * & 0 & * & 0 & 0 & * & 0 & 0 \end{array} \right]$$

Contained in form (ns1):

(ns3) associated to the weight $((1, 1, 0, -2), (1, 0, -1), (1, 0, -1))$:

$$\left[\begin{array}{ccc|ccc|ccc} * & * & * & * & * & * & * & 0 & 0 \\ * & * & * & * & * & * & * & 0 & 0 \\ * & * & 0 & * & * & 0 & * & 0 & 0 \end{array} \right]$$

(ns4) associated to the weight $((2, 0, -1, -1), (1, 0, -1), (1, 0, -1))$:

$$\left[\begin{array}{ccc|ccc|ccc} * & * & * & * & * & * & * & * & 0 \\ * & * & * & * & * & 0 & * & 0 & 0 \\ * & * & * & * & 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

Contained in unstable form (u2):

(ns_5) associated to the weight $((0, 0, 0, 0), (2, -1, -1), (1, 1, -2))$:

$$\left[\begin{array}{ccc|ccc|ccc|ccc} * & * & * & * & * & * & * & * & * & * & * & * \\ * & * & * & * & * & * & * & * & * & * & * & * \\ * & 0 & 0 & * & 0 & 0 & * & 0 & 0 & * & 0 & 0 \end{array} \right]$$

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